MERIT ENERGY COMPANY

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June 26, 2025

Kansas Department of Health & Environment
Bureau of Air
1000 SW Jackson, Suite 310
Topeka, Kansas 66612-1366

**RE: Hugoton Compressor Station**
**Stevens County, Kansas**
**Source ID: 1890015**
**Notification of Compliance Status for U-164**

To Whom It May Concern:

In accordance with 40 CFR 63.6645(h), Merit Energy Company, LLC (Merit) is submitting this Notification of Compliance Status following the completion of a performance test for one Waukesha L7042 GSI 1,232 horsepower compressor engine, U-164, at the Hugoton Compressor Station (Facility). The semi-annual performance test was completed on May 15, 2025. The information required by 40 CFR 63.9(h)(2)(i) is listed below.

A. The methods that were used to determine compliance:
   ***The methods that were used to determine compliance are detailed in Section 3.0 of the attached performance test report.***

B. The results of any performance tests, opacity or visible emission observations, continuous monitoring system (CMS) performance evaluations, and/or other monitoring procedures or methods that were conducted:
   ***The results of the performance test are attached.***

C. The methods that will be used for determining continuing compliance, including a description of monitoring and reporting requirements and test methods:
   ***Continuous compliance will be determined by monitoring the catalyst inlet temperature and monthly monitoring of the pressure drop across the catalyst in accordance with Table 6 of 40 CFR 63, Subpart ZZZZ. Semi-annual performance tests will also be completed.***

D. The type and quantity of hazardous air pollutants emitted by the source (or surrogate pollutants if specified in the relevant standard), reported in units and averaging times and in accordance with the test methods specified in the relevant standard:
   ***U-164 emits formaldehyde. See the results of the attached performance test.***

E. If the relevant standard applies to both major and area sources, an analysis demonstrating whether the affected source is a major source (using the emissions data generated for this notification):

   ***Hugoton Compressor Station has a potential to emit greater than 25 tons per year (TPY) for total hazardous air pollutants (HAPs) and greater than 10 TPY for a single HAP; therefore, Hugoton Compressor Station is a major source.***

F. A description of the air pollution control equipment (or method) for each emission point, including each control device (or method) for each hazardous air pollutant and the control efficiency (percent) for each control device (or method):

   ***U-164 is equipped with a non-selective catalyst which limits the formaldehyde to 350 ppbvd or less at 15 percent oxygen. The catalyst specification sheet is attached.***

G. A statement by the owner or operator of the affected existing, new, or reconstructed source as to whether the source has complied with the relevant standard or other requirements:

   ***The statement that the affected source has complied with 40 CFR 63, Subpart ZZZZ is attached.***

H. A written report of the average percent load determination:

   ***In accordance with 40 CFR 63.6620(i), the average percent load determination is included in the attached performance test report.***

If you have any questions regarding this notification or require additional information, please contact me at bailey.sharp@meritenergy.com or at (972) 628-1454.

Sincerely,

MERIT ENERGY COMPANY

Bailey Sharp

Bailey Sharp
Environmental Analyst

## Certification of Truth, Accuracy and Completeness

### 40 CFR 63.9(h)(2)(i)(G)

Based on information and belief formed after reasonable inquiry, I certify that the statements and information contained in this notification of compliance status are true, accurate, and complete to the best of my knowledge and U-164 at Hugoton Compressor Station has complied with the applicable requirements of 40 CFR 63, Subpart ZZZZ.

Name (typed): Chad Brister, Senior Vice President

(Signed): ____________________ Date: 6/26/25

RJ Mann & Associates Inc.
Engine Controls & Panels/Compressor Parts

MIRATECH

# Equipment Specification

**Proposal Information**

| | | | |
|---|---|---|---|
| Proposal Number: | RJM-22-007577 | Date: | **12/7/2022** |
| Project Reference: | Waukesha 7042 | | |

**Engine Information**

| | | | |
|---|---|---|---|
| Engine Make: | Waukesha | Speed: | Rated |
| Engine Model: | 7042 | Power Output: | 896 bhp |
| Rated Speed: | 1000 RPM | Exhaust Flow Rate: | 5,153 acfm (cfm) |
| Fuel Description: | Natural Gas | Exhaust Temperature: | 1,060 ° F |
| Hours Of Operation: | 8760 Hours per year | $O_2$: | 0.3% |
| Load: | 100% | $H_2O$: | 18.5% |

**Emission Data (100% Load)**

| Emission | Raw Engine Emissions | | | | | | Target Outlet Emissions | | | | | | Calculated Reduction |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | *g/bhp-hr* | *tons/yr* | *ppmvd @ 15% $O_2$* | *ppmvd* | *g/kW-hr* | *lb/MW-hr* | *g/bhp-hr* | *tons/yr* | *ppmvd @ 15% $O_2$* | *ppmvd* | *g/kW-hr* | *lb/MW-hr* | |
| $NO_x$* | *13* | 112.48 | 704 | 2,456 | 17.433 | 38.43 | *4* | 34.61 | 216 | 756 | 5.364 | 11.83 | 69.2% |
| CO | *9* | 77.87 | 800 | 2,793 | 12.069 | 26.61 | *4* | 34.61 | 356 | 1,241 | 5.364 | 11.83 | 55.6% |

**System Specifications**

**Catalyst (Replacement Catalyst)**

| | |
|---|---|
| Element Model Number: | MECB-TW-RO-2650-0000-350 |
| Number of Catalyst Layers: | 1 |
| Number of Catalyst Per Layer: | 1 |
| Catalyst Back Pressure: | 3.0 inWC (Clean) |
| Design Exhaust Flow Rate: | 5,153 acfm |
| Design Exhaust Temperature: | 1,060f |
| Dimensions: | Ø 26.5 in |
| Exhaust Temperature Limits**: | 750f – 1250f (catalyst inlet); 1350f (catalyst outlet) |
| System Pressure Loss: | 3.0 inWC (Clean) |

* MW referenced as $NO_2$

** General catalyst temperature operating range. Performance is based on the Design Exhaust Temperature.

GAS®

303 West 3rd Street • Elk City, Oklahoma 73644 • 580-225-0403

May 28, 2025

Kansas Department of Health and Environment
1000 SW Jackson Street
Topeka, KS 66612

RE: Merit Energy Company
40 CFR Part 63 Subpart ZZZZ – Test Report Submittal

To Whom It May Concern:

We are notifying you of a performance testing report for 40 CFR Part 63 Subpart ZZZZ on behalf of Merit Energy. Please see below for the information pertinent to the testing. Testing took place the day of May 15, 2025.

| **Facility/Unit** | **Unit ID** | **Serial Number** | **Location** |
|---|---|---|---|
| Hugoton Compressor Station | APC-164 | PP168 | Stevens County, Kansas |

For information regarding the performance test, you can contact me by email at klackey@gasinc.us or by phone at 580-225-0403. For information regarding engine or location information, please contact Bailey Sharp by email at bailey.sharp@meritenergy.com or by phone at 972-628-1454.

Regards,

Katie Lackey

Katie Lackey
Client Services Development Supervisor
GAS Inc.

**40 CFR Part 63 Subpart ZZZZ**

**Performance Test Report**
**Test Type: Semiannual**

**Test Date: 5/15/2025**

**DOM:** 1/1/75

**Source:**
Waukesha L7042GSI

Rich Burn (4 Cycle)

**Unit Number:** APC-164

**Serial Number:** PP168

**Engine Hours:** 18818

**Permit #:** #1890015

**Location:**
Hugoton Compressor Station

Stevens County, Kansas

**Prepared on Behalf of:**
Merit Energy Company

| | | | | CH2O |
|---|---|---|---|---|
| | | | | |
| Results ppmvd | | | | 0.001 |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |

Test Started: 10:40 AM    Test Completed: 02:40 PM

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303 W. 3rd St    (580) 225-0403    Elk City , OK 73644

# Index



## Tables



## Figures



## Appendices

## 1.0 Key Personnel

| | |
|---|---|
| GAS | Hunter Newton |
| Merit Energy Company | Stacy Burrows |

## 2.0 Sampling System

The sampling system used consisted of a Stainless steel probe, heated Teflon line, gas conditioning system, and a Gasmet model DX4000 FTIR analyzer. The gas conditioning system used was a Gasmet Personal Sampling System with a Zirconium Oxide oxygen sensor.

## 3.0 Methods Used

### ASTM D6348-03

This extractive FTIR based field test method is used to quantify gas phase concentrations of multiple target analytes (CO, NOX, CH2O, & VOC's) from stationary source effluent. Because an FTIR analyzer is potentially capable of analyzing hundreds of compounds, this test method is not analyte or source specific. The analytes, detection levels, and data quality objectives are expected to change for any particular testing situation. It is the responsibility of the tester to define the target analytes, the associated detection limits for those analytes in the particular source effluent, and the required data quality objectives for each specific test program. Provisions are included in this test method that require the tester to determine critical sampling system and instrument operational parameters, and for the conduct of QA/QC procedures. Testers following this test method will generate data that will allow an independent observer to verify the valid collection, identification, and quantification of the subject target analytes.

### EPA Method 1 & 1A

The purpose of the method is to provide guidance for the selection of sampling ports and traverse points at which sampling for air pollutants will be performed pursuant to regulations set forth in this part.

### EPA Method 2 & 2C

This method is applicable for the determination of the average velocity and the volumetric flow rate of a gas stream. The average gas velocity in a stack is determined from the gas density and from measurement of the average velocity head with a standard pitot tube. Velocity readings are taken from each stack at 16 separate traverse points (Table 6.1) and used to determine the engines mass emissions rate, calculated utilizing the formulas seen in section 7.0 of this report.

### EPA Method 3A

This is a procedure for measuring oxygen (O2) and carbon dioxide (CO2) in stationary source emissions using a continuous instrumental analyzer. Quality assurance and quality control requirements are included to assure that the tester collects data of known quality. Documentation to these specific requirements for equipment, supplies, sample collection and analysis, calculations, and data analysis will be included.

## 4.0 Test Summary

Unit APC-164 with a serial number of PP168 which is a Waukesha L7042GSI engine located at Hugoton Compressor Station and operated by Merit Energy Company was tested for emissions of: Formaldehyde. The test was conducted on 5/15/2025 by Hunter Newton with Great Plains Analytical Services, Inc. All quality assurance and quality control tests were within acceptable tolerances.

The engine is a natural gas fired Rich Burn (4 Cycle) engine rated at 1232 brake horse power (BHP) at 1000 RPM. The engine was operating at 1107 BHP and 898 RPM which is 89.83% of maximum engine load during the test. The test HP calculation can be found on page 8. The engine was running at the maximum load available at the test site.

This test will satisfy the testing requirements for 40 CFR Part 63 Subpart ZZZZ. Unit APC-164 is authorized to operate under permit #1890015.

## Site Verification Photos

## 4.0 Test Summary

| Engine/Compressor Specs | | | |
|---|---|---|---|
| Location | Hugoton Compressor Station | Unit ID | APC-164 |
| Make | Waukesha | Site Elevation ft. | 3100 |
| Model | L7042GSI | Atmospheric Pressure psi. | 13.15 |
| Serial Number | PP168 | Stack Diameter in. | 12.5 |
| mfg. rated hp | 1232 | Catalyst | Yes |
| mfg. rated rpm | 1000 | *Date of Manufacture* | 1/1/1975 |

| Engine/Compressor Operation | Run 1 | Run 2 | Run 3 | 3 Run Averages |
|---|---|---|---|---|
| Test Horsepower | 1122 | 1099 | 1099 | 1107 |
| Test RPM | 911 | 892 | 892 | 898 |
| Percent Load % | 91.10% | 89.20% | 89.20% | 89.83% |
| Intake Manifold Pressure (hg) | 47.00 | 46.00 | 48.00 | 47.00 |
| Intake Manifold Temperature (F) | 157.00 | 155.00 | 150.00 | 154.00 |
| **Ambient Conditions** | | | | |
| Ambient Temperature Dry (F) | 68.00 | 74.00 | 76.00 | 72.67 |
| **Exhaust Flow Data** | | | | |
| Q Stack (dscfh) | 72785.85 | 72249.10 | 72147.26 | 72394.07 |
| Q Stack (dscm/hr) | 2061.05 | 2045.85 | 2042.96 | 2049.95 |
| Moisture Fraction Bws | 0.18 | 0.18 | 0.18 | 0.18 |
| **Results** | | | | |
| Method 3A Corrected O2% Dry | 0.00% | 0.00% | 0.00% | 0.00% |
| Moisture % | 18.28% | 18.16% | 18.11% | 18.18% |

Test Summary

| Permitted Standards | | | | Results | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | 10:40 AM | 11:56 AM | 01:07 PM | | 5/15/2025 |
| | **Federal** | | **ZZZZ** | **Run 1** | **Run 2** | **Run 3** | **3 Run Average** | **Pass Permits** |
| CH2O (ppmvd) @15% O2 | | | | 0.001 | 0.000 | 0.000 | 0.000 | |
| CH2O (ppbvd) @15% O2 | | | 350.000 | 0.691 | 0.282 | 0.000 | 0.324 | Pass |
| CH2O (ppmvw) | | | | 0.002 | 0.001 | 0.000 | 0.001 | |
| CH2O (ppmvd) | | | | 0.002 | 0.001 | 0.000 | 0.001 | |
| CH2O (mol wt) 30.03 | | | | | | | | |

## 5.0 Run Summaries

**Table 5.1 Run Summaries**

Summary Source Run 1

| PPM Wet | | | | | | CH2O | Oxygen % | Moisture % |
|---|---|---|---|---|---|---|---|---|
| Baseline | | | | | | 0.03 | 0.00% | 0.00% |
| Source | | | | | | 0.00 | 0.00% | 18.28% |
| Spike | | | | | | 0.00 | 0.00% | 16.30% |
| Source | | | | | | 0.00 | 0.00% | 18.29% |
| Baseline | | | | | | 0.04 | 0.00% | 0.00% |
| Avg. Wet | | | | | | 0.00 | 0.00% | 18.28% |

Summary Source Run 2

| PPM Wet | | | | | | CH2O | Oxygen % | Moisture % |
|---|---|---|---|---|---|---|---|---|
| Baseline | | | | | | 0.04 | 0.00% | 0.00% |
| Source | | | | | | 0.00 | 0.00% | 18.17% |
| Spike | | | | | | 0.00 | 0.00% | 16.67% |
| Source | | | | | | 0.00 | 0.00% | 18.15% |
| Baseline | | | | | | 0.03 | 0.00% | 0.02% |
| Avg. Wet | | | | | | 0.00 | 0.00% | 18.16% |

Summary Source Run 3

| PPM Wet | | | | | | CH2O | Oxygen % | Moisture % |
|---|---|---|---|---|---|---|---|---|
| Baseline | | | | | | 0.03 | 0.00% | 0.02% |
| Source | | | | | | 0.00 | 0.00% | 18.11% |
| Spike | | | | | | 0.00 | 0.00% | 16.99% |
| Source | | | | | | 0.00 | 0.00% | 18.11% |
| Baseline | | | | | | 0.06 | 0.00% | 0.00% |
| Avg. Wet | | | | | | 0.00 | 0.00% | 18.11% |

Run Summaries

GAS

## 6.0 Volumetric Flow Rate Data



### Table 6.1. Data used for volumetric flow rate (Method 2)

Volumetric Flow Rates

| Volumetric Flow Rate Data | | | | | |
|---|---|---|---|---|---|
| **Pitot Tube Coefficient Cp(std)=** .99 | | | | | |
| **Stack diameter =** 12.5 | **or:** 1.04 **feet** | **or:** .85 **Square feet** | | | |
| | | Run 1 | Run 2 | Run 3 | Average |
| H2O | % | 18.28 | 18.16 | 18.11 | 18.18 |
| CO2 | %d | 11.31 | 11.42 | 11.24 | 11.32 |
| O2 | %d | 0.00 | 0.00 | 0.00 | 0.00 |
| CO | %d | 0.44 | 0.46 | 0.46 | 0.46 |
| Molecular Weight Stack Gas dry basis (Md) | g/g mole | 29.81 | 29.96 | 29.93 | 29.94 |
| Molecular Weight Stack Gas wet basis (Ms) | g/g mole | 27.65 | 27.79 | 27.77 | 27.77 |
| Stack Static Pressure (Pg) | "H20 | 0.45 | 0.45 | 0.49 | 0.46 |
| Stack Static Pressure (Pg) | "Hg | 0.03 | 0.03 | 0.04 | 0.03 |
| Atmospheric Pressure at Location (Pbar) | MBAR | 1010.16 | 1006.44 | 1009.48 | 1008.69 |
| Atmospheric Pressure at Location (Pbar) | "Hg | 29.84 | 29.73 | 29.82 | 29.80 |
| Absolute Stack Pressure (Ps) | "Hg | 29.87 | 29.76 | 29.86 | 29.83 |
| Stack Temperature | Deg C | 441.22 | 445.14 | 448.99 | 445.12 |
| Stack Temperature | Deg F | 826.19 | 833.25 | 840.19 | 833.21 |
| Stack Temperature | Deg R | 1285.86 | 1292.92 | 1299.86 | 1292.88 |
| Stack Gas Velocity | ft/sec | 70.84 | 71.28 | 73.89 | 71.97 |
| Stack Flow Rate Q | cfs | 60.36 | 60.37 | 60.37 | 60.37 |
| Stack Gas Wet Volmetric Flow Rate | scf/hr | 89070.63 | 88278.76 | 88102.65 | 88484.01 |
| Stack Gas Dry Volumetric Flow Rate | scf/hr | 72785.85 | 72249.10 | 72147.26 | 72394.07 |

| Emissions Sampling Points - 3 point long line sampling probe | Inches |
|---|---|
| First Sampling Point Taken @ 16.7% of Stack Diameter | 2.09 |
| Second Sampling Point Taken @ 50% of Stack Diameter | 6.25 |
| Third Sampling Point Taken @ 83.3% of Stack Diameter | 10.41 |

### Table 6.2. Stack gas pressure readings measured with a standard pitot tube use for Volumetric Flow Rate

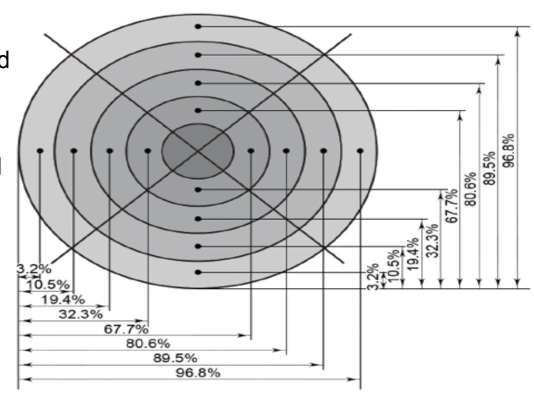
| Δpstd = Velocity head measured by the standard pitot tube, (in.) H2O. | **Run 1** | | **Run 2** | | **Run 3** | |
|---|---|---|---|---|---|---|
| | **Δpstd (in) H2O** | **Exhaust Temp F** | **Δpstd (in) H2O** | **Exhaust Temp F** | **Δpstd (in) H2O** | **Exhaust Temp F** |
| Pitot Tube Sampling Points (Velocity) | | | | | | |
| 1 | 0.31 | 826.00 | 0.61 | 833.00 | 0.37 | 840.00 |
| 2 | 0.30 | 827.00 | 0.55 | 831.00 | 0.44 | 841.00 |
| 3 | 0.28 | 828.00 | 0.53 | 835.00 | 0.48 | 840.00 |
| 4 | 0.33 | 824.00 | 0.51 | 833.00 | 0.41 | 838.00 |
| 5 | 0.36 | 825.00 | 0.39 | 832.00 | 0.52 | 832.00 |
| 6 | 0.38 | 826.00 | 0.33 | 834.00 | 0.48 | 839.00 |
| 7 | 0.41 | 824.00 | 0.48 | 833.00 | 0.60 | 841.00 |
| 8 | 0.48 | 827.00 | 0.46 | 835.00 | 0.61 | 844.00 |
| 9 | 0.55 | 828.00 | 0.49 | 836.00 | 0.62 | 845.00 |
| 10 | 0.63 | 827.00 | 0.51 | 833.00 | 0.57 | 832.00 |
| 11 | 0.67 | 825.00 | 0.47 | 832.00 | 0.55 | 844.00 |
| 12 | 0.60 | 826.00 | 0.44 | 833.00 | 0.51 | 845.00 |
| 13 | 0.55 | 828.00 | 0.43 | 831.00 | 0.48 | 841.00 |
| 14 | 0.50 | 827.00 | 0.38 | 835.00 | 0.43 | 842.00 |
| 15 | 0.42 | 826.00 | 0.36 | 832.00 | 0.38 | 840.00 |
| 16 | 0.36 | 825.00 | 0.32 | 834.00 | 0.33 | 839.00 |
| Δpstd Average= | 0.45 | 826.19 | 0.45 | 833.25 | 0.49 | 840.19 |

| | |
|---|---|
| Sample after Back Purge: | 0.33 |
| Within 5% of Last Δpstd reading: | Yes |

| | |
|---|---|
| Stack Diameter (inches) | 12.50 |
| Inches upstream from disturbance | 10.00 |
| Inches downstream from disturbance | 23.00 |

**Figure 6.1**

16 Traverse Points Were Used




Pitot readings are taken for Method 2 calculations using measuring points outlined in Method 1

***The exhaust stack did not present cyclonic flow conditions at the sampling location due to the absence of cyclones, inertial demisters, venturi scrubbers, or tangential inlets.**

*** Cyclonic Flow Check (Pass/Fail): PASS**

| Strat Check | |
|---|---|
| Check 1 | 279.737 |
| Check 2 | 279.111 |
| Check 3 | 283.953 |
| Gas Used | methane |
| Pass/Fail | Pass |

# 7.0 Calculations

## Method 2: Determination of Stack Gas Velocity and Volumetric Flow Rate

*Note- Use of this method negates the need for any fuel related numbers for emissions calculations

### Nomenclature

Δp(avg) = Velocity head of stack gas, mm H2O (in. H2O).

3600 = Conversion Factor, sec/hr.

A = Cross-sectional area of stack, m2 (ft2).

Bws = Water vapor in the gas stream (from ASTM D6348)

Cp(std) = Standard pitot tube coefficient; use 0.99

Kp = Velocity equation constant.

Md = Molecular weight of stack gas, dry basis, g/g-mole (lb./lb.-mole).

Ms = Molecular weight of stack gas, wet basis, g/g-mole (lb./lb. -mole).

Ps = Absolute stack pressure (Pbar+ Pg), mm Hg (in Hg)

Pstd = Standard absolute pressure, 760 mm Hg (29.92 in. Hg).

Qsd = Dry volumetric stack gas flow rate corrected to standard conditions, dscm/hr. (dscf/hr.).

Ts(abs) = Absolute stack temperature, °K (°R). = 460 + Ts for English units.

Tstd = Standard absolute temperature, 293°K (528 °R).

Vs = Average stack gas velocity, m/sec (ft./sec).

| *(Observed from Method 3 12.3) Dry Molecular Weight. Equation 3-1 | |
|---|---|
| Md = .44(%CO2)+.32(%O2)+.28(%N2+%CO) | Md = .44(11.31)+ .32(0)+ .28(88.245+.445) = 29.81 LB/LB-MOLE |
| **12.5 Molecular Weight of Stack Gas. Equation 2-6** | |
| Ms = Md(1-Bws)+18.0(Bws) | Ms = 29.81(1 -.18283)+ 18.0(.18283) = 27.65 LB/LB-MOLE |
| **12.6 Average Stack Gas Velocity. Equation 2-7** | |
| Vs = Kp*Cp(std)*√Δpavg*√(Ts(abs)/(Ps*Ms)) | Vs = 85.49*.99*√.45*√(1285.86/(29.87*27.65)) = 70.84 FT/SEC |
| **12.7 Average Stack Gas Volumetric Flow Rate. Equation 2-8** | |
| Qsd = 3600(1-Bws)Vs*A((Tstd*Ps)/(Ts(abs)*Pstd)) | Qsd = 3600(1-.18283)70.84*.852((528*29.87)/ (1285.86*29.92)) = 72785.85 DSCF/HR |
| ***Standard conversion from feet to meters** | |
| Q = Qsd/35.315 | Q = 72785.847/35.315 = 2061.046 DSCM/HR |

## Emission Rates (Examples use CO Run 1)

### Nomenclature

453.6= Conversion factor lb. to gram

A = Cross-sectional area of stack, m2 (ft2).

BHP/HR. = Brake work of the engine, horsepower-hour (HP-HR.).

BTU/HP-HR. = Brake Specific Fuel Consumption (HHV)

ER = Emission rate of (CO) in g/HP-hr.

F(d )= Volumes of combustion components per unit of heat content, scm/J (scf/million Btu).

Q = Stack gas volumetric flow rate, in standard cubic meters per hour, dry basis

LB/HR.= Emission rate of (CO) in LB/HR.

Mol wt.= Mol Weight of CO (28.01)

HP= Engines rated Horsepower

Mfg.= Manufacturer Exhaust flow rate at 100% (ft3/min)

O2 = Concentration of oxygen on a dry basis, percent.

ppm= Parts Per Million (CO)

ppm@15% O2= PPM corrected to 15% O2

Qsd = Dry volumetric stack gas flow rate corrected to standard conditions, dscm/hr. (DSCF/HR.).

Run Time = Run Time in Minutes

TPY= Tons per Year

Vs = Average stack gas velocity, m/sec (ft./sec).

| CO corrected @ 15% O2 | |
|---|---|
| ppm @ 15% O2 = PPM*((20.9-15%O2)/(20.9-O2)) | ppm @ 15% O2 = 4445.606*(20.9-15)/(20.9-0%)) = 1254.98 PPM @ 15% O2 |
| **G/HP HR from 40 CFR Part 60 Subpart JJJJ** | |
| g/hp-hr = (PPM*(1.164*10^-3)*Q*(Run Time/60))/BHP/HR | (4445.606*(1.164*10^-3)*2061.046*(Run Time/60))/1122.352 = 9.503 G/HP-HR |
| **LB/HR from Grams** | |
| lb/hr = ER*1/453.6*BHP-HR | LB/HR = 9.503*1/453.6*1122.352 = 23.512 LB/HR |
| **TPY** | |
| TPY = LB/HR. *4.38 | TPY = 23.512*4.38 = 102.983 TPY |
| **Horsepower** | |
| | HP Provided On Site |
| **Wet to Dry PPM** | |
| ppm wet * (1/(1-H2O) = ppm dry | 4445.61 = 3632.8155*(1/(1-.18)) |

### 8.0 Oxygen Calibration

**8.1 Calibration error test; how do I confirm my analyzer calibration is correct?** After the tester has assembled, prepared and calibrated the sampling system and analyzer, they conduct a 3-point analyzer calibration error test before the first run and again after any failed system bias test or failed drift test. They then introduce the low-, mid-, and high-level calibration gases sequentially in direct calibration mode. At each calibration gas level (low, mid, and high) the calibration error must be within ± 2.0 percent of the calibration span.

**8.2 Initial system bias and system calibration error checks.** Before sampling begins, it is determined whether the high- level or mid-level calibration gas best approximates the emissions and it is used as the upscale gas. The upscale gas is introduced at the probe upstream of all sample-conditioning components in system calibration mode.

(1) Next, zero gas is introduced as described above. The response must be within 0.5 percent of the upscale gas concentration.
(2) Low-level gas reading is observed until it has reached a final, stable value and the results are recorded. The measurement system will be operated at the normal sampling rate during all system bias checks.
(3) If the initial system bias specification is not met, corrective action is taken. The applicable calibration error test from Section 8.2.3 of EPA Method 7E is repeated along with the initial system bias check until acceptable results are achieved, after which sampling will begin. The pre- and post-run system bias must be within ± 5.0 percent of the calibration span for the low-level and upscale calibration gases.

**8.3 Post-run system bias check and drift assessment - confirming that each sample collected is valid.** Sampling may be performed for multiple runs before performing the post-run bias or system calibration error check provided this test is passed at the conclusion of the group of runs. A failed final test in this case will invalidate all runs subsequent to the last passed test.

(1) If the post-run system bias check is not passed, then the run is invalid. The problem is then diagnosed and fixed, then another calibration error test and system bias is passed before repeating the run.
(2) After each run, the low-level and upscale drift is calculated, using Equation 7E–4 in Section 12.5 from EPA Method 7E. If the post-run low- and upscale bias checks are passed, but the low-or upscale drift exceeds the specification in Section 13.3, the run data are valid, but a 3-point calibration error test and a system bias check must be performed and passed prior to additional testing.

**Table 8.1 Oxygen Calibration**

**Method 7E 3.4** To the extent practicable, the measured emissions should be between 20 to 100 percent of the selected calibration span. This may not be practicable in some cases of low concentration measurements or testing for compliance with an emission limit when emissions are substantially less than the limit.

**EPA Method 3A QA Worksheet**

| Certified Gas Concentraion Low-Level (%) | Certified Gas Concentraion Mid-Level (%) | Certified Gas Concentraion High-Level (%) |
|---|---|---|
| 0.00% | 10.59% | 20.92% |

**(DIRECT) Analyzer Calibration Error (≤ 2%)**

**Linearity Check**

| | Certified Concentration Value (%) | Direct Calibration Response (%) | Absolute Difference (%) | Analyzer Calibration Error (%) |
|---|---|---|---|---|
| Zero Gas % | 0.00% | 0.00% | 0.00% | 0.00% |
| Mid-Level Gas % | 10.59% | 10.52% | 0.07% | 0.33% |
| High-Level Gas % | 20.92% | 20.80% | 0.12% | 0.59% |

7E 8.5: Note: that you may risk sampling for multiple runs before performing the post-run bias provided you pass this test at the conclusion of the group of runs

**(SYSTEM) Calibration Bias Checks (≤ 5%) and Drift Checks (≤ 3%)** — Upscale Gas 10.59%

| | |
|---|---|
| Zero Offset | 0.00% |
| Span | 20.92 |

| | Analyzer Calibration Response (%) | Bias Pre Initial Value: System Calibrations Response Pre (%) | Bias Pre Initial Value: System Bias (% of Span) Pre | Bias Post Inital Values: System Calibration Response Post (%) | Bias Post Inital Values: System Bias (% of Span) Post | Drift (% of Span) |
|---|---|---|---|---|---|---|
| Zero Gas | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| Upscale Gas | 10.52% | 10.70% | 0.86% | 10.52% | 0.01% | 0.01% |

**(SYSTEM) Calibration Bias Checks (≤ 5%) and Drift Checks (≤ 3%)**

| | | | |
|---|---|---|---|
| Avg. Gas Concentration (Run 1) | 0.00% | Effluent Gas (Cgas) Run 1 | 0.00% |
| Avg. Gas Concentration (Run 2) | 0.00% | Effluent Gas (Cgas) Run 2 | 0.00% |
| Avg. Gas Concentration (Run 3) | 0.00% | Effluent Gas (Cgas) Run 3 | 0.00% |

**EPA Method 3A QA Worksheet**

| | |
|---|---|
| Zero Gas | 100% Nitrogen |
| Mid-Level Gas | 10.59% |
| High-Level Gas | 20.92% |

**O2 QA/QC**

| | Analyzer Direct Calibration Response | Certified bottle value % |
|---|---|---|
| Zero Gas % | 0.00% | 0.00% |
| Mid-Level Gas % | 10.52% | 10.59% |
| High-Level Gas % | 20.80% | 20.92% |

| | System Calibration Response Pre (%) | |
|---|---|---|
| Zero Gas % | 0.00% | Upscale Used |
| Upscale Cal | 10.70% | 10.59% |

| | System Calibration Response Post (%) | |
|---|---|---|
| Zero Gas % | 0.00% | Upscale Used |
| Upscale Cal | 10.52% | 10.59% |

## 9.0 Engine Parameter Data Sheet

Great Plains Analytical Services, Inc

| | |
|---|---|
| **Company** | Merit Energy Company |
| **Facility** | Hugoton Compressor Station |
| **Date** | 5/15/2025 |
| **Site Elevation (ft)** | 3100 |
| **Unit ID** | APC-164 |
| **Make** | Waukesha |
| **Model** | L7042GSI |
| **Serial Number** | PP168 |
| **Technician** | Hunter Newton |

| | **Run 1** | **Run 2** | **Run 3** | **Completed** |
|---|---|---|---|---|
| **Run Start Times** | 10:40 AM | 11:56 AM | 01:07 PM | 02:40 PM |
| **Engine Hours** | 18815 | 18816 | 18817 | 18818 |

| | **Engine Parameter Data** | | | |
|---|---|---|---|---|
| | Run 1 | Run 2 | Run 3 | Average |
| **Engine Speed (RPM)** | 911.0 | 892.0 | 892.0 | 898.3 |
| **Intake Manifold Pressure (psi)** | 47.0 | 46.0 | 48.0 | 47.0 |
| **Intake Manifold Temp °F** | 157.0 | 155.0 | 150.0 | 154.0 |
| **Engine Load (BHP)** | 1122.4 | 1098.9 | 1098.9 | 1106.7 |
| **Ambient Temp °F** | 68.0 | 74.0 | 76.0 | 72.7 |
| **Humidity %** | 26.0 | 18.0 | 17.0 | 20.3 |
| **Dew Point °F** | 32.0 | 27.0 | 27.0 | 28.7 |
| **AFR Manufacturer/Type** | Altronic | Altronic | Altronic | Altronic |
| | | | | |
| | | | | |
| **Suction Pressure (psi)** | 27.0 | 27.0 | 27.0 | 27.0 |
| **Discharge Pressure (psi)** | 183.0 | 360.0 | 360.0 | 301.0 |
| **Catalyst (Yes or No)** | Yes | | | |
| **Catalyst Manufacturer** | Miratech | Miratech | Miratech | Miratech |
| **# of Catalyst Installed** | 1 | 1 | 1 | 1 |
| **Catalyst Inlet Temp °F** | 867.0 | 828.0 | 828.0 | 841.0 |
| **Catalyst Outlet Temp °F** | 812.0 | 811.0 | 811.0 | 811.3 |
| **Catalyst Pressure Drop H2O** | 0.0 | 0.0 | 0.0 | 0.0 |

## 10.0 QA/QC Results

| Pre-test Baseline Results | Time: |
|---|---|
| Carbon monoxide CO | Average: 0.05 |
| Nitrogen monoxide NO | Average: 0.10 |
| Nitrogen dioxide NO2 | Average: 0.02 |
| NOx | Average: 0.12 |
| VOC | Average: 0.03 |
| Oxygen | Average: 0.00 |

| CTS Direct to Analyzer | Time: |
|---|---|
| CTS Bottle Concentration | Value: 99.57 |
| CTS Compound Concentration Avg | Value: 101.09 |
| Tolerance | 2.00% |
| Difference between measured and expected | 1.52% |

| Pre-test System Zero | Time: |
|---|---|
| Carbon monoxide CO | Average: 0.00 |
| Nitrogen monoxide NO | Average: 0.00 |
| Nitrogen dioxide NO2 | Average: 0.08 |
| NOx | Average: 0.08 |
| VOC | Average: 0.06 |
| Oxygen | Average: 0.00 |

QA/QC Results

| Pre-test System CTS & Mechanical Response Time | |
|---|---|
| Mechanical Response Time | 45 seconds |
| CTS Bottle Concentration | Value: 99.57 |
| CTS Compound Concentration Avg | Value: 100.70 |
| Tolerance | 5.00% |
| Difference between measured and expected | 1.13% |

| System (Equilibration) Response Time Target (Spike) | |
|---|---|
| Equilibration Response Time | 45 seconds |
| Spike Reported | Value: 253.14 |
| Spike Expected | Value: 249.80 |

| System (Zero) Response Time | |
|---|---|
| System Zero Response Time | 45 seconds |
| System Response Time | 45 seconds |

## 11.0 D6348 Annexes 1-8

### D6348 Annexes 1. Test Plan Requirements

### Annex 1.2

The test quality objectives completed for the emissions test are demonstrated throughout Annexes 1, 2, 3, 4, 5, 6, 7 & 8 as layed out per ASTM D6348-03. All reference methods, pre-test and post test procedures were within acceptable limits. Data generated during the pre-test and post-test procedures are summarized below in order of the distinctive Annex.

Three 60 minute test runs were performed. The final analyte concentrations are the average of each test run. Data was taken at 60 second intervals. Each 60 second measurement was the average of 600 scans.

Acetaldehyde is used as the surrogate compound for the Annex 5 Spiking Technique due to Acetaldehyde being the closest spectral analysis match to Formaldehyde found in the combustion process of natural gas.

**Annex Table 1.2.1 Certified Calibration Bottle Concentrations**

| Bottle | Expiration | NO2 | Ethylene | SF6 | Acetaldehyde | O2 (%) |
|---|---|---|---|---|---|---|
| CC476212 | 8/26/2025 | | | 10.19 | 100.3 | |
| CC502013 | 2/12/2028 | 98.74 | | | | 10.59% |
| EB0115896 | 12/18/2027 | | 99.57 | | | |
| CC440196 | 7/13/2032 | | | | | 20.92% |

| Cylinder # CC333629 Expiration: 10-18-2032 | | | | |
|---|---|---|---|---|
| | **Propane** | **CO** | **NO** | **SF6** |
| Bottle Value | 249.80 | 501.50 | 254.60 | 9.99 |
| Analyzer System Response | 253.14 | 510.59 | 251.89 | 9.73 |
| Percent Difference | 1.34% | 1.81% | 1.07% | 2.57% |

**Annex Table 1.2.2 Measurement System Capabilities**

| Parameter Measured | Gas | Concentration (ppm) | Path Length | Equilibration Time | Dilution Factor | % Recovery |
|---|---|---|---|---|---|---|
| Path Length | Ethylene | 101.086 | 5.076 | | | |
| 5-Gas Spike Direct | Propane | 251.144 | | | | |
| | SF6 | 9.932 | | | | |
| | CO | 503.972 | | | | |
| | NO | 250.528 | | | | |
| 2-Gas Spike Direct | Acetaldehyde | 99.725 | | | | |
| | SF6 | 10.200 | | | | |
| Mechanical Response Time | Ethylene | 100.698 | | 43 seconds | | |
| 5-Gas Analyzer Response | Propane | 253.138 | | 17 seconds | | |
| | SF6 | 9.731 | | | | |
| | CO | 510.586 | | | | |
| | NO | 251.887 | | | | |
| 2-Gas Analyzer Response | Acetaldehyde | 97.753 | | 36 seconds | | |
| | SF6 | 10.155 | | | | |
| Analyte Spike Recovery | Acetaldehyde & SF6 | | | | 9.70% | 110.06% |
| | | | | | 6.26% | 123.17% |
| | | | | | 4.62% | 129.41% |
| System Zero | Nitrogen | | | 39 seconds | | |
| Post Spike System | Propane | 252.231 | | | | |
| | CO | 498.558 | | | | |
| | NO | 249.775 | | | | |
| | SF6 | 9.692 | | | | |

## Annex 1.3

**Annex Table 1.3.1 Test Specific Target Analytes and Data Quality Objectives**

| Compounds | Infrared Analysis Region (cm-1) | Expected Concentration Range | Measurement System Achievable Minimum Detectable Concentrations | Required Measurement System Accuracy and Precision for Test Application |
|---|---|---|---|---|
| CO | 2000-2200 | 0-1200 ppm | 0.16267 ppm | 4 ppm |
| NO | 1875-2138 | 0-1000 ppm | 0.4007 ppm | 2 ppm |
| NO2 | 2700-2950 | 0-100 ppm | 0.4899 ppm | 2 ppm |
| VOC | 2600-3200<br>910-1150<br>2550-2950 | 0-100 ppm | 1.8520 ppm Total VOC's | 1 ppm per VOC |
| CH2O | 2550-2850 | 0-100 ppm | 0.7878 ppm | 1 ppm |
| **Interfering Compounds** | * CO is analyzed in a separate analysis region than CO2 and H2O | | | |
| CO2 | 926-1150 | 0-10% | 0% | n/a |
| Water Vapor | 3200-3401 | 0-22% | 0.20% | n/a |

* VOCs compiled of Acetaldehyde, Ethylene, Hexane, and Propane.

## Annex 1.4

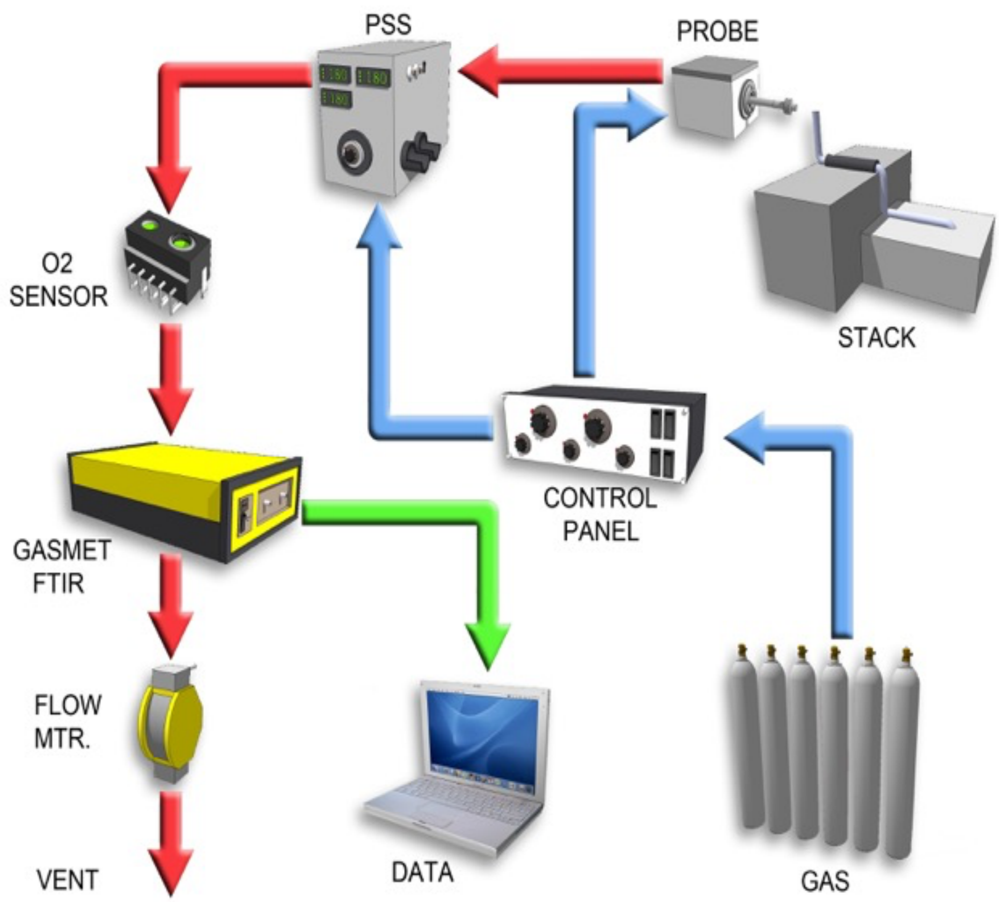
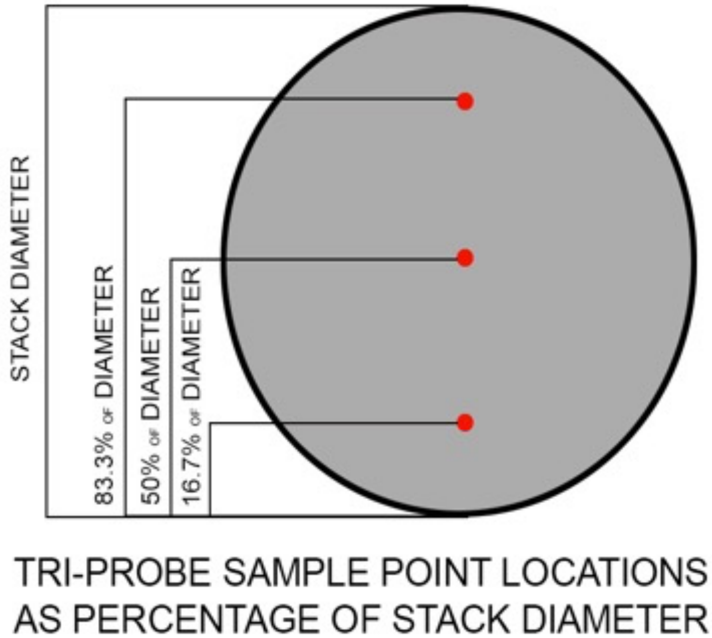



**Figure Annex 1.4.1 Sampling Train**

The testing instrumentation is housed in an enclosed vehicle which is located approximately 45 feet from the source. A heated sample line (sixty feet in length) is attached to the inlet of analyzer system and the source effluent discharges through the FTIR outlet.




| Sampling Point Locations in Inches | |
|---|---|
| 16.7% | 2.088 |
| 50% | 6.250 |
| 83.3% | 10.413 |

**Figure Annex 1.4.2 Sampling Points**

Interior Stack Diameter (inches): 12.5

Inches Upstream: 10

Inches Downstream: 23

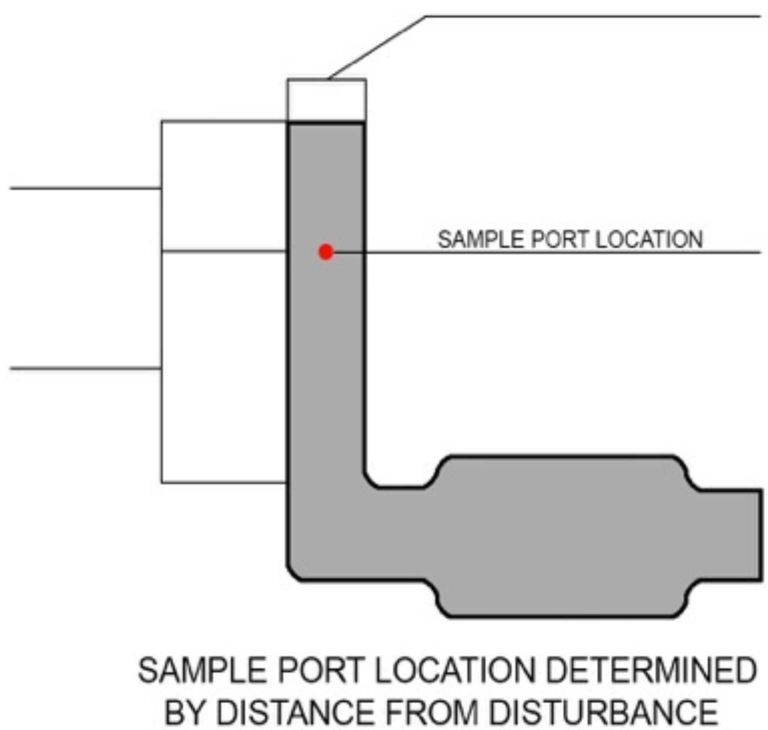



**Figure Annex 1.4.3 Sample Port Location**

| | |
|---|---|
| Pressure @ Sampling Point (Pg "H2O): | 0.46 |
| Temperature @ Sampling Point (Deg F): | 833.21 |
| Stack Gas Dry Volumetric Flow Rate | 72394.07 |
| H2O%d @ Sampling Point: | 18.18 |
| CO2%d @ Sampling Point: | 11.32 |

GAS

## D6348 Annex 2. Determination of FTIR Measurement System Minimum Detectable Concentrations (MDC#1)

| Target Analyte | Results (ppm) |
|---|---|
| CO | 0.1627 |
| NO | 0.4007 |
| NO2 | 0.4899 |
| Ethylene | 0.3782 |
| Propane | 0.4351 |
| Hexane | 0.2233 |
| Acetaldehyde | 0.8152 |
| Formaldehyde | 0.7878 |

$$NEA^{m}_{rms} = \sqrt{\frac{1}{n}\sum_{j=1}^{N_m}\left(NEA^{m}_{i}\right)^2}$$

$$REF^{m}_{rms} = \sqrt{\frac{1}{n}\sum_{j=1}^{N_m}\left(REF^{m}_{i}\right)^2}$$

$$MDC\#1 = \frac{NEA^{m}_{rms}}{REF^{m}_{rms}} * \frac{C_{ref}L_{ref}}{L_{cell}}$$

## D6348 Annex 3. FTIR Reference Spectra

| Calibration Transfer Standard | Expected | Measured | Path Length | Validated |
|---|---|---|---|---|
| Ethylene | 99.57 | 101.09 | 5.076 | Passed |

## D6348 Annex 4. Required Pre-Test Procedures

**Annex Table 1.2.2 Measurement System Capabilities**

| Parameter Measured | Gas | Concentration (ppm) | Path Length | Equilibration Time | Dilution Factor | % Recovery |
|---|---|---|---|---|---|---|
| Path Length | Ethylene | 101.086 | 5.076 | | | |
| 5-Gas Spike Direct | Propane | 251.144 | | | | |
| | SF6 | 9.932 | | | | |
| | CO | 503.972 | | | | |
| | NO | 250.528 | | | | |
| 2-Gas Spike Direct | Acetaldehyde | 99.725 | | | | |
| | SF6 | 10.200 | | | | |
| Mechanical Response Time | Ethylene | 100.698 | | 43 seconds | | |
| 5-Gas Analyzer Response | Propane | 253.138 | | 17 seconds | | |
| | SF6 | 9.731 | | | | |
| | CO | 510.586 | | | | |
| | NO | 251.887 | | | | |
| 2-Gas Analyzer Response | Acetaldehyde | 0.000 | | 36 seconds | | |
| | SF6 | 10.155 | | | | |
| Analyte Spike Recovery | Acetaldehyde & SF6 | | | | 9.70% | 110.06% |
| | | | | | 6.26% | 123.17% |
| | | | | | 4.62% | 129.41% |
| System Zero | Nitrogen | | | 39 seconds | | |
| Post Spike System | Propane | 252.231 | | | | |
| | CO | 498.558 | | | | |
| | NO | 249.775 | | | | |
| | SF6 | 9.692 | | | | |

## D6348 Annex 5. Analyte Spiking Technique

| Parameter | Gas | Concentration | Measured | % Difference | Specification | Validated |
|---|---|---|---|---|---|---|
| Spike Direct | Propane | 249.800 | 251.144 | 0.54% | +/- 2% | Pass |
| | SF6 | 9.988 | 9.932 | 0.56% | +/- 2% | Pass |
| | CO | 501.500 | 503.972 | 0.49% | +/- 2% | Pass |
| | NO | 254.600 | 250.528 | 1.60% | +/- 2% | Pass |
| | | | | | | |
| | Acetaldehyde | 100.300 | 99.725 | 0.57% | +/- 2% | Pass |

**Spike Run 1 via the System**

| | Source Output | Spike Average | Dilution Factor | Expected | % Recovery | Specification |
|---|---|---|---|---|---|---|
| Acetaldehyde | 1.256 | 11.956 | | 10.863 | 110.060% | 70-130% |
| SF6 | 0.000 | 0.989 | 9.958% | | | <10% |

**Spike Run 2 via the System**

| | Source Output | Spike Average | Dilution Factor | Expected | % Recovery | Specification |
|---|---|---|---|---|---|---|
| Acetaldehyde | 1.125 | 9.033 | | 7.334 | 123.170% | 70-130% |
| SF6 | 0.000 | 0.639 | 6.434% | | | <10% |

**Spike Run 3 via the System**

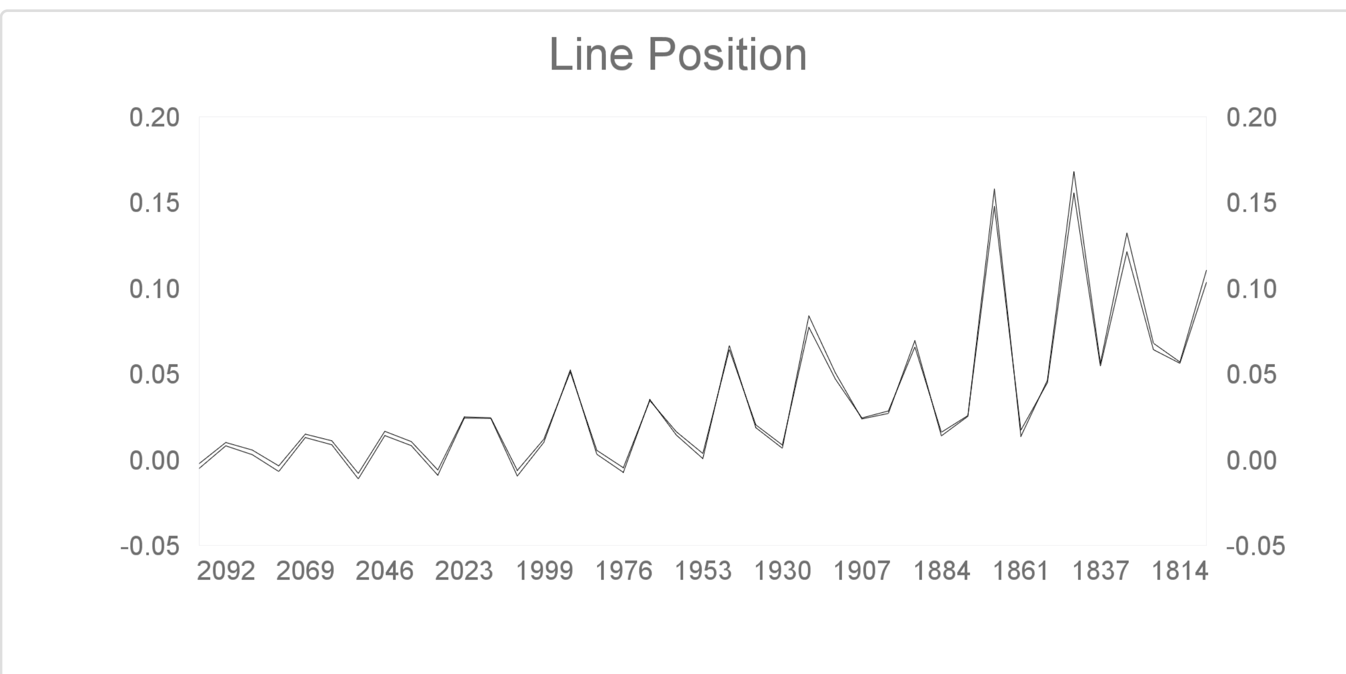
| | Source Output | Spike Average | Dilution Factor | Expected | % Recovery | Specification |
|---|---|---|---|---|---|---|
| Acetaldehyde | 1.145 | 7.410 | | 5.726 | 129.410% | 70-130% |
| SF6 | 0.000 | 0.471 | 4.742% | | | <10% |

## D6348 Annex 6. Determination of System Performance Parameters

**Noise Equivalent Absorbance (NEA)**

RMS High 0.000340
RMS Mid 0.000199
RMS Low 0.000297

**Line Position: Demonstrating the peak positions align correctly (Manual Comparison +/- 20%)**



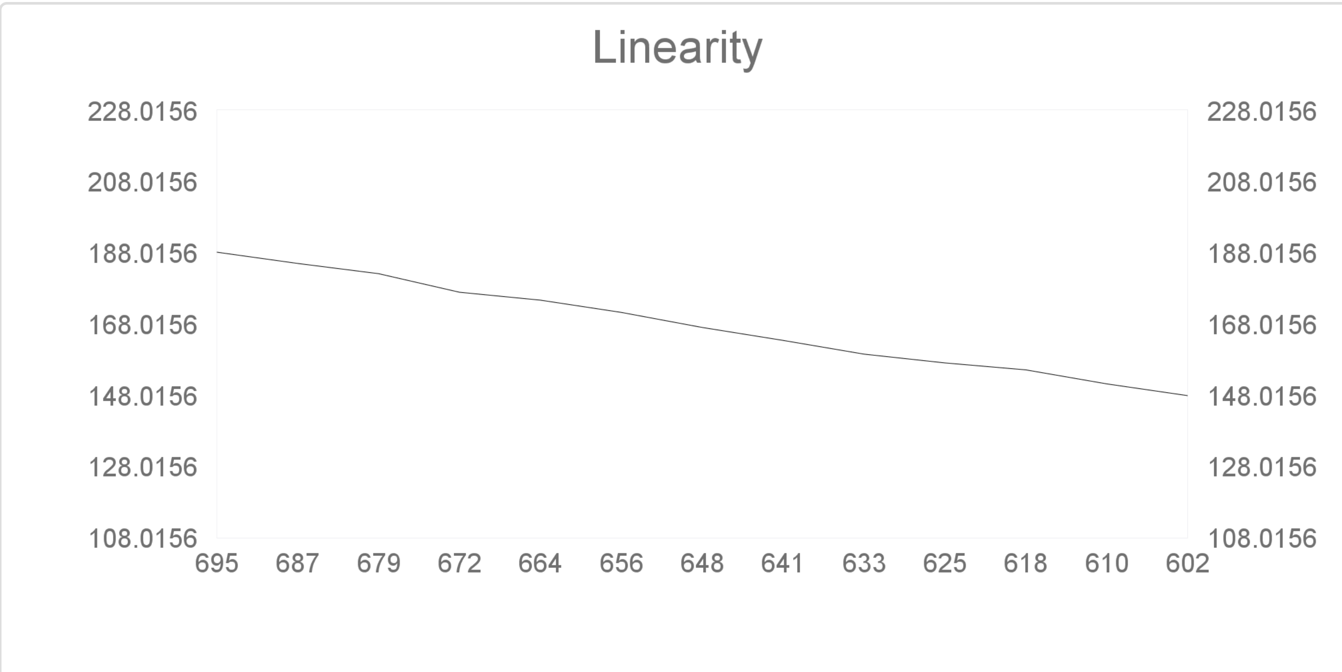

| Line Position Difference: | 0.00% | Pass | Resolution Difference to Reference | 0.53% | Pass |
|---|---|---|---|---|---|

## Resolution

The Gasmet GICCOR (Genzel Interferometer with Cube Corner Retroreflectors) interferometer is specially designed for maximum optical throughput and maximum signal to noise ratio of 7.72 (cm-1) remaining stable with any vibration and temperature changes.

## Detector Linearity

The Gasmet DX4000 is a low resolution spectrometer where the aperture is fixed to a maximum angle setting and the detector linearity was tested with an alternate approach.  A three point linearity of the CTS gas was performed and validated.




| | | |
|---|---|---|
| Regression Slope: | 40.6540179 | Pass |

## D6348 Annex 7. Preparation of Analytical Quantification Algorithm

The analytical accuracy of the quantification algorithm is satisfied via the results from Annex 5 per Annex 7.6

## D6348 Annex 8. Post Test Quality Assurance/Quality Control Procedures

| POST CTS System Check: | |
|---|---|
| CTS Bottle Concentration: | 99.57 |
| CTS Sample Concentration Average: | 99.77 |
| Difference between measured and expected: | 0.20% |
| Tolerance: | 5.00% |

| Run Data Validation - Automated vs Manual | Reading | Validated | Comments |
|---|---|---|---|
| Run 1 Points 1 & 2 on CO/NO/Propane | All within 20% | Passed | Demonstrates no interferences observed. |
| Run 2 Points 1 & 2 on CO/NO/Propane | All within 20% | Passed | Demonstrates no interferences observed. |
| Run 3 Points 1 & 2 on CO/NO/Propane | All within 20% | Passed | Demonstrates no interferences observed. |

## 12.0 Signature Page

**Job/File Name:** Merit Energy Company; Hugoton Compressor Station; APC-164; ZZZZ

We certify that based on review of test data, knowledge of those individuals directly responsible for conducting this test, we believe the submitted information to be accurate and complete.

| | | | |
|---|---|---|---|
| Company: | G.A.S. Inc. | Date: | 5/15/2025 |
| Print Name: | Travis Hartley | | |
| Title: | Director of Stack Testing | | |
| Signature: | | | |
| Phone Number: | 580-225-0403 | | |

| | | | |
|---|---|---|---|
| Company: | G.A.S. Inc. | Date: | 5/15/2025 |
| Print Name: | Hunter Newton | | |
| Title: | Emissions Specialist | | |

| | | | |
|---|---|---|---|
| Company: | | | |
| Print Name: | | Date: | |
| Signature: | | | |
| Title: | | | |
| Phone Number: | | | |

# Appendices

Spike (5 Gas)

**Airgas Specialty Gases**
Airgas USA LLC
12722 S. Wentworth Ave.
Chicago,IL 60628
Airgas.com

# CERTIFICATE OF ANALYSIS

## Grade of Product: EPA PROTOCOL STANDARD

| | | | |
|---|---|---|---|
| Part Number: | E05NI94E15AC014 | Reference No: | 54-403166330-1 |
| Cylinder Number: | CC333629 | Cylinder Volume: | 147.0 |
| Laboratory: | 124 - Chicago (SAP) - IL | Cylinder Pressure: | 2015 |
| PGVP Number: | B12024 | Valve Outlet: | 660 |
| Gas Code: | CO,CO2,NO,NOX,PPN,BALN | Certification Date: | Oct 18, 2024 |

**Expiration Date: Oct 18, 2032**

Certification performed in accordance with "EPA Traceability Protocol for Assay and Certification of Gaseous Calibration Standards (May 2012)" document EPA 600/R-12/531, using the assay procedures listed. Analytical Methodology does not require correction for analytical interference. This cylinder has a total analytical uncertainty as stated below with a confidence level of 95%. There are no significant impurities which affect the use of this calibration mixture. All concentrations are on a mole/mole basis unless otherwise noted. The results relate only to the items tested. The report shall not be reproduced except in full without approval of the laboratory. Do Not Use This Cylinder below 100 psig, i.e. 0.7 megapascals.

### ANALYTICAL RESULTS

| Component | Requested Concentration | Actual Concentration | Protocol Method | Total Relative Uncertainty | Absolute Uncertainty | Assay Dates |
|---|---|---|---|---|---|---|
| NOX | 250.0 PPM | 255.4 PPM | G1 | +/- 1.4% NIST Traceable | +/- 3.6PPM NIST Traceable | 10/18/2024 10/18/2024 |
| NITRIC OXIDE | 250.0 PPM | 254.6 PPM | G1 | +/- 1.4% NIST Traceable | +/- 3.6PPM NIST Traceable | 10/18/2024 10/18/2024 |
| PROPANE | 250.0 PPM | 249.8 PPM | G1 | +/- 0.6% NIST Traceable | +/- 1.6PPM NIST Traceable | 10/18/2024 |
| CARBON MONOXIDE | 500.0 PPM | 501.5 PPM | G1 | +/- 0.4% NIST Traceable | +/- 1.8PPM NIST Traceable | 10/18/2024 |
| CARBON DIOXIDE | 5.000 % | 4.980 % | G1 | +/- 1.1% NIST Traceable | +/- 0.1% NIST Traceable | 10/18/2024 |
| NITROGEN | Balance | | | - | - | |

### CALIBRATION STANDARDS

| Type | Lot ID | Cyl No | Concentration | Uncertainty | Exp Date |
|---|---|---|---|---|---|
| GMIS | DCK1128202232 | CC754457 | 249.1 PPM NITRIC OXIDE NITROGEN | +/- 0.9% | Jan 18, 2031 |
| RGM | 12425 | EB0112909 | 250.5 PPM NITRIC OXIDE NITROGEN | +/- 0.8% | Oct 19, 2026 |
| PRIMARY | 122-402992913-1 | CC507051 | 4.22 PPM NITROGEN DIOXIDE AIR | +/- 10% | Mar 21, 2027 |
| SRM | 1680b | FF20776 | 494.8 PPM CARBON MONOXIDE NITROGEN | +/- 0.2% | Feb 07, 2028 |
| NTRM | 20060203 | 6162649Y | 243.3 PPM PROPANE NITROGEN | +/- 0.5% | Mar 17, 2027 |
| GMIS | 160402954567106 | CC301815 | 497.6 PPM CARBON MONOXIDE NITROGEN | +/- 0.3% | Apr 04, 2032 |
| NTRM | 13060426 | CC413698 | 7.489 % CARBON DIOXIDE NITROGEN | +/- 0.6% | May 14, 2025 |

The SRM, NTRM, PRM, or RGM noted above is only in reference to the GMIS used in the assay and not part of the analysis.

### ANALYTICAL EQUIPMENT

| Instrument/Make/Model | Analytical Principle | Last Multipoint Calibration |
|---|---|---|
| Nicolet iS50 AUP2010242 | FTIR | Oct 03, 2024 |
| CO-1 SIEMENS ULTRAMAT 6E N1J5700 | NDIR | Oct 03, 2024 |
| Nicolet iS50 AUP2010242 | FTIR | Oct 03, 2024 |
| Nicolet iS50 AUP2010242 | FTIR | Oct 03, 2024 |
| Nicolet iS50 AUP2110277 | FTIR | Oct 03, 2024 |

### NON-EPA ANALYTICAL RESULTS

| Component | Requested/Max Concentration | Concentration |
|---|---|---|
| SULFUR HEXAFLUORIDE | 10.00 PPM | 9.988 PPM |

**Triad Data Available Upon Request**

Signature on File

Approved for Release



GAS

**Airgas Specialty Gases**
Airgas USA LLC
525 North Industrial Loop Road
Tooele, UT 84074
Airgas.com

# CERTIFICATE OF ANALYSIS

## Grade of Product: EPA PROTOCOL STANDARD

| | | | |
|---|---|---|---|
| Part Number: | E03NI89E15W0020 | Reference Number: | 153-403252877-1 |
| Cylinder Number: | CC502013 | Cylinder Volume: | 145.3 CF |
| Laboratory: | 124 - Tooele (SAP) - UT | Cylinder Pressure: | 2015 PSIG |
| PGVP Number: | B72025 | Valve Outlet: | 660 |
| Gas Code: | NO2,O2,BALN | Certification Date: | Feb 12, 2025 |

**Expiration Date: Feb 12, 2028**

Certification performed in accordance with "EPA Traceability Protocol for Assay and Certification of Gaseous Calibration Standards (May 2012)" document EPA 600/R-12/531, using the assay procedures listed. Analytical Methodology does not require correction for analytical interference. This cylinder has a total analytical uncertainty as stated below with a confidence level of 95%. There are no significant impurities which affect the use of this calibration mixture. All concentrations are on a mole/mole basis unless otherwise noted. The results relate only to the items tested. The report shall not be reproduced except in full without approval of the laboratory. Do Not Use This Cylinder below 100 psig, i.e. 0.7 megapascals.

### ANALYTICAL RESULTS

| Component | Requested Concentration | Actual Concentration | Protocol Method | Total Relative Uncertainty | Assay Dates |
|---|---|---|---|---|---|
| NITROGEN DIOXIDE | 100.0 PPM | 98.74 PPM | G1 | +/- 2.0% NIST Traceable | 02/05/2025, 02/12/2025 |
| OXYGEN | 10.60 % | 10.59 % | G1 | +/- 0.7% NIST Traceable | 02/05/2025 |
| NITROGEN | Balance | | | | |

### CALIBRATION STANDARDS

| Type | Lot ID | Cylinder No | Concentration | Uncertainty | Expiration Date |
|---|---|---|---|---|---|
| GMIS | 1534002024510 | CC511408 | 59.65 PPM NITROGEN DIOXIDE/NITROGEN | 1.8% | Apr 04, 2027 |
| PRM | 12438 | D153564 | 59.5 PPM NITROGEN DIOXIDE/AIR | 1.7% | Nov 21, 2024 |
| NTRM | 11060608 | CC338459 | 14.93 % OXYGEN/NITROGEN | 0.2% | Nov 02, 2028 |

The SRM, NTRM, PRM, or RGM noted above is only in reference to the GMIS used in the assay and not part of the analysis.

### ANALYTICAL EQUIPMENT

| Instrument/Make/Model | Analytical Principle | Last Multipoint Calibration |
|---|---|---|
| MKS FTIR NO2 018143349 | FTIR | Jan 16, 2025 |
| Horiba MPA-510 W603MM58 O2 | O2 Paramagnetic (DIXON) | Jan 09, 2025 |

**Triad Data Available Upon Request**

Signature on file
**Approved for Release**

Airgas
an Air Liquide company

**Airgas Specialty Gases**
Airgas USA LLC
12722 S. Wentworth Ave.
Chicago, IL 60628
Airgas.com

# CERTIFICATE OF ANALYSIS

## Grade of Product: PRIMARY STANDARD

| | | | |
|---|---|---|---|
| Part Number: | X02NI99P15ACVH8 | Reference Number: | 54-403224032-1 |
| Cylinder Number: | EB0115896 | Cylinder Volume: | 144.0 CF |
| Laboratory: | 124 - Chicago (SAP) - IL | Cylinder Pressure: | 2015 PSIG |
| Analysis Date: | Dec 18, 2024 | Valve Outlet: | 350 |
| Lot Number: | 54-403224032-1 | | |

**Expiration Date: Dec 18, 2027**

Primary Standard Gas Mixtures are traceable to N.I.S.T. weights and/or N.I.S.T. Gas Mixture reference materials.

## ANALYTICAL RESULTS

| Component | Req Conc | Actual Concentration (Mole %) | Analytical Uncertainty |
|---|---|---|---|
| ETHYLENE | 100.0 PPM | 99.57 PPM | +/- 1% |
| NITROGEN | Balance | | |

Signature on file
**Approved for Release**

**Airgas Mid South region**
Airgas USA LLC
9741 E. 56th St. North
Tulsa, OK 74117
Airgas.com

# CERTIFICATE OF ANALYSIS

## Grade of Product: ULTRA HIGH PURITY-PURE

| | | | |
|---|---|---|---|
| Part Number: | NI UHP300 | Reference Number: | 29-403219707-1 |
| Cylinder Number: | 4641953Y | Cylinder Volume: | 304.0 CF |
| Laboratory: | 106 - Tulsa Fast Fill (SAP) - OK | Cylinder Pressure: | 2640 PSIG |
| Analysis Date: | Dec 13, 2024 | Valve Outlet: | 580 |
| Lot Number: | 29-403219707-1 | | |

## ANALYTICAL RESULTS

| Component | Requested Purity | Certified Concentration |
|---|---|---|
| NITROGEN | 99.999 % | 99.999 % |
| THC | 0.5 PPM | 0.23 PPM |
| CO + CO2 | 1.0 PPM | 0.1 PPM |
| Moisture | 1.0 PPM | 0.709 PPM |
| Oxygen | 1.0 PPM | 0.96 PPM |

Impurities verified against analytical standards traceable to NIST by weight and/or analysis.

Signature on file
**Approved for Release**

21% O2

**Airgas**
an Air Liquide company

**Airgas Specialty Gases**
Airgas USA LLC
12722 S. Wentworth Ave.
Chicago, IL 60628
Airgas.com

# CERTIFICATE OF ANALYSIS

## Grade of Product: EPA PROTOCOL STANDARD

| | | | |
|---|---|---|---|
| Part Number: | E02NI79E15A00B1 | Reference Number: | 54-403116776-1 |
| Cylinder Number: | CC440196 | Cylinder Volume: | 146.0 CF |
| Laboratory: | 124 - Chicago (SAP) - IL | Cylinder Pressure: | 2015 PSIG |
| PGVP Number: | B12024 | Valve Outlet: | 590 |
| Gas Code: | O2,BALN | Certification Date: | Aug 13, 2024 |

**Expiration Date: Aug 13, 2032**

Certification performed in accordance with "EPA Traceability Protocol for Assay and Certification of Gaseous Calibration Standards (May 2012)" document EPA 600/R-12/531, using the assay procedures listed. Analytical Methodology does not require correction for analytical interference. This cylinder has a total analytical uncertainty as stated below with a confidence level of 95%. There are no significant impurities which affect the use of this calibration mixture. All concentrations are on a mole/mole basis unless otherwise noted. The results relate only to the items tested. The report shall not be reproduced except in full without approval of the laboratory. Do Not Use This Cylinder below 100 psig, i.e. 0.7 megapascals.

### ANALYTICAL RESULTS

| Component | Requested Concentration | Actual Concentration | Protocol Method | Total Relative Uncertainty | Assay Dates |
|---|---|---|---|---|---|
| OXYGEN | 21.00 % | 20.92 % | G1 | +/- 1.4% NIST Traceable | 08/13/2024 |
| NITROGEN | Balance | | | | |

### CALIBRATION STANDARDS

| Type | Lot ID | Cylinder No | Concentration | Uncertainty | Expiration Date |
|---|---|---|---|---|---|
| NTRM | 08010217 | K011889 | 23.20 % OXYGEN/NITROGEN | +/- 0.4% | Feb 16, 2030 |

### ANALYTICAL EQUIPMENT

| Instrument/Make/Model | Analytical Principle | Last Multipoint Calibration |
|---|---|---|
| O2-1 HORIBA MPA-510 W2TP7BN5 | PARAMAGNETIC | Aug 07, 2024 |

**Triad Data Available Upon Request**

Signature on file
**Approved for Release**



GAS

**Airgas Specialty Gases**
Airgas USA LLC
9810 BAY AREA BLVD
Pasadena, TX 77507
Airgas.com

# CERTIFICATE OF ANALYSIS

## Grade of Product: PRIMARY STANDARD

| | | | |
|---|---|---|---|
| Part Number: | X03NI99P15ACD42 | Reference Number: | 163-403122990-1 |
| Cylinder Number: | CC476212 | Cylinder Volume: | 140.0 CF |
| Laboratory: | 124 - Pasadena (SG06) - TX | Cylinder Pressure: | 2015 PSIG |
| Analysis Date: | Aug 26, 2024 | Valve Outlet: | 350SS |
| Lot Number: | 163-403122990-1 | | |

**Expiration Date: Aug 26, 2025**

Primary Standard Gas Mixtures are traceable to N.I.S.T. weights and/or N.I.S.T. Gas Mixture reference materials.

## ANALYTICAL RESULTS

| Component | Req Conc | Actual Concentration (Mole %) | Analytical Uncertainty |
|---|---|---|---|
| SULFUR HEXAFLUORIDE | 10.00 PPM | 10.19 PPM | +/- 5% |
| ACETALDEHYDE | 100.0 PPM | 100.3 PPM | +/- 2% |
| NITROGEN | Balance | | |

Signature on file
**Approved for Release**

Tri Probe Certification

**Great Plains Analytical Services**
303 W 3rd St
Elk City, OK, 73644
(580)225-0403 Fax: (580)225-2612

# CERTIFICATE OF ANALYSIS

**Grade of Product: CERTIFIED STANDARD-PROBE**

| | | | |
|---|---|---|---|
| Part Number: | 12L | Reference Number: | 25 |
| Laboratory: | GAS INC. | Stack Diameter: | 12" |
| Analysis Date: | 6/14/2022 | Target Flow Rate: | 3L/min |
| LOT Number: | A | Number of Points: | 3 |
| SN: | 25A 12L | | |

Product performance verified by direct comparison to calibration standards traceable to N.I.S.T. *The probe listed on this form meets the multipoint traverse requirement of EPA Method 7e, section 8.4 as shown in the accompanying data. Method 7e, section 8.4 states that the multipoint traverse requirement can be satisfied by sampling via "a multi-hole probe designed to sample at the prescribed points with a flow +/- 10 percent of mean flow rate".

## ANALYTICAL RESULTS

| Run # | Total Flow (L/m) | Measured Flow Port A (L/m) (Delta q1) | Measured Flow Port B (L/m) (Delta q2) | Measured Flow Port C (L/m) (Delta q3) | Mean Probe Port Sampled Flow (L/m) |
|---|---|---|---|---|---|
| Run 1 | 2 LPM | 0.710 (5.97△) | 0.670 (0.00△) | 0.630 (-5.97△) | 0.670 |
| Run 2 | 4 LPM | 1.40 (5.79△) | 1.32 (-0.25△) | 1.25 (-5.54△) | 1.32 |

*Calibration conducted in accordance with Emission Measurement Center Guideline Document – **EMC GD-031**

Notes:

Approved for Release

6/14/2022

Date

Probe size: 12L

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY
RESEARCH TRIANGLE PARK, NC 27711

OFFICE OF
AIR QUALITY PLANNING
AND STANDARDS

March 15, 2021

Mr. Jordan Williamson
CEO
GAS Inc.
303 W. 3rd Street
Elk City, OK 73644

Dear Mr. Williamson:

We are writing in response to your letter received on September 17, 2020, in which you request the approval of alternative testing procedures. The EPA's Office of Air Quality Planning and Standards (OAQPS) is the delegated authority for consideration of major alternatives to test methods and procedures as set forth in 40 CFR parts 60 and 63 under which your request must be addressed. GAS Inc. is requesting a change to one of the test methods, ASTM D6348-03, used for conducting performance tests to determine compliance under 40 CFR part 60, Subpart JJJJ – Standards of Performance for Stationary Spark Ignition Internal Combustion Engines (Subpart JJJJ) and 40 CFR part 63, Subpart ZZZZ – National Emissions Standards for Hazardous Air Pollutants for Performance for Stationary Reciprocating Internal Combustion Engines (Subpart ZZZZ). The change being requested will be used to check detector linearity of the Fourier Transform Infrared (FTIR) instrumentation used to conduct this method. Specifically, you are requesting that the procedures of section 8.3.3 of Method 320 (40 CFR part 60, Appendix A), another FTIR-based method allowed under Subparts JJJJ and ZZZZ, be used in lieu of section A6.4.1 of ASTM D6348-03 when conducting testing using ASTM D6348-03 under 40 CFR part 60, Subpart JJJJ and 40 CFR part 63, Subpart ZZZZ.

In your request, you state that this alternative linearity check procedure will produce consistent results when utilizing either Method 320 or ASTM D6348-03. Additionally, some FTIR instrumentation does not allow for reducing the size of the aperture in the instrument and, thus, it would not be feasible to properly conduct the entirety of the ASTM D6348-03 method in its current form using such an instrument.

Based on our understanding of FTIR instrument principles and recognition that the requested alternative determination of detector linearity is both technically sound and contained within Method 320, we are approving the requested change. We believe that this alternative is acceptable for use for use in testing all engines subject to 40 CFR part 60 Subpart JJJJ and 40 CFR part 63, Subpart ZZZZ. Also, we will post this letter as ALT-141 on EPA's website (at *www.epa.gov/emc/broadly-applicable-approved-alternative-testmethods*) to announce that our approval of this alternative test method is broadly applicable to engines for the purposes of meeting Subparts JJJJ and ZZZZ.

If you should have any questions or require further information regarding this approval, please contact David Nash of my staff at 919-541-9425 or email at *nash.dave@epa.gov*.

Sincerely,

STEFFAN JOHNSON

Digitally signed by STEFFAN JOHNSON
Date: 2021.03.15 12:28:50 -04'00'

Steffan M. Johnson, Group Leader
Measurement Technology Group

cc:

Sara Ayers, EPA/OECA/OC/MAMPD, (ayres.sara@epa.gov)
Melanie King, EPA/OAR/OAQPS/SPPD, (king.melanie@epa.gov)
James Leather, EPA Region 6, (leather.james@epa.gov)
David Nash, EPA/OAR/OAQPS/AQAD, (nash.dave@epa.gov)

**Hunter Newton**

**GAS**
**580-225-0403**
**info@gasinc.us**

***Type of Sources Tested:***

Stationary Internal Combustion Engines
- 4 Stroke Rich Burn Engines
- 2 Stroke & 4 Stroke Lean Burn Engines

Stationary Natural Gas Fired Generators

Stationary Propane Fired Generators

Gas Fired Boilers

***Types of Analyzers:***

- Gasmet DX4000 FTIR
- Gasmet Portable Sampling Unit with Zirconium Oxide O2 Sensory
- Testo 350
- Flame Ionization Detector

***Qualifications:***

Trained, studied, and fully demonstrates compliance for emissions testing via data collection outlined in the following Reference Methods:
- EPA Method 1 & 1A – Sampling & Traverse Points
- EPA Method 2 & 2C – Velocity & Volumetric Flow Rate of a Gas Stream
- EPA Method 3A – Oxygen
- EPA Method 7E – NOX
- EPA Method 10 – Carbon Monoxide
- EPA Method 25A – Volatile Organic Compounds
- ASTM D6348 – Extractive Fourier Transform Infrared Spectroscopy

Conducts emissions testing on a weekly basis including, but not limited to, the following test types: Initial Compliance, Biennial Compliance, Semiannual Compliance & Quarterly Compliance. All tests performed are in accordance to any and all Federal & State requirements as applicable (i.e. JJJJ, ZZZZ, 106.512, 117, PEA, etc.). Performed testing in Colorado, Utah, Wyoming, North Dakota, Montana, Kansas, New Mexico, Oklahoma, Texas, Louisiana (land and off-shore), Arkansas, Ohio, Pennsylvania, West Virginia, New York, Kentucky, & Mississippi.

- Quarterly Performance Reviews covering ongoing changes with Federal Regulations, State Compliance guidelines, & site-specific safety certifications.

# RAW Result Data

| 1 Minute Average Data Sheet | | | |
|---|---|---|---|
| **Client:** | Merit Energy Company | **Make:** | Waukesha |
| **Location:** | Hugoton Compressor Station | **Model:** | L7042GSI |
| **Unit Number:** | APC-164 | **Serial Number:** | PP168 |

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Stability** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 06:45 | 0.040 | 1.610 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 11.984 | 274.961 | 0.000 | 0.000 | 0.774 | 0.000 | 1.521 | 0.000 | 2.104 | 0.657 | 0.000 | 0.00 | 1.96 | 928.0 | 0.00 | 180.0 |
| **Pre Test Baseline Direct** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 06:53 | -0.000 | 0.010 | 0.032 | 0.000 | 0.000 | 0.055 | 0.001 | 0.033 | 0.000 | 0.042 | 0.136 | 0.000 | 0.024 | 0.016 | 0.005 | 0.049 | 0.000 | 0.002 | 0.00 | 0.05 | 927.0 | 0.00 | 180.0 |
| 5/15/2025 | 06:54 | -0.000 | 0.002 | 0.083 | 0.296 | 0.000 | 0.000 | 0.009 | 0.000 | 0.000 | 0.002 | 0.000 | 0.012 | 0.000 | 0.000 | 0.005 | 0.000 | 0.035 | 0.000 | 0.30 | 0.03 | 927.0 | 0.00 | 180.0 |
| 5/15/2025 | 06:54 | -0.002 | 0.001 | 0.045 | 0.000 | 0.072 | 0.000 | 0.005 | 0.004 | 0.025 | 0.000 | 0.063 | 0.000 | 0.000 | 0.000 | 0.001 | 0.000 | 0.000 | 0.001 | 0.07 | 0.00 | 928.0 | 0.00 | 180.0 |
| **CTS Direct** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 06:59 | 0.011 | 0.027 | 0.000 | 0.913 | 0.000 | 0.000 | 0.000 | 0.132 | 0.000 | 101.370 | 0.027 | 0.000 | 0.238 | 0.000 | 0.017 | 0.004 | 0.096 | 0.018 | 0.91 | 67.68 | 926.0 | 0.00 | 180.0 |
| 5/15/2025 | 06:59 | 0.011 | 0.000 | 0.000 | 0.840 | 0.000 | 0.000 | 0.000 | 0.106 | 0.000 | 101.086 | 0.000 | 0.000 | 0.248 | 0.000 | 0.022 | 0.041 | 0.093 | 0.022 | 0.84 | 67.50 | 925.0 | 0.00 | 180.0 |
| 5/15/2025 | 07:00 | 0.010 | 0.000 | 0.000 | 1.009 | 0.000 | 0.000 | 0.006 | 0.279 | 0.000 | 100.801 | 0.000 | 0.026 | 0.151 | 0.000 | 0.015 | 0.027 | 0.000 | 0.021 | 1.01 | 67.23 | 926.0 | 0.00 | 180.0 |
| **High O2 Direct** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 07:06 | -0.000 | 0.006 | 0.000 | 0.000 | 0.182 | 0.159 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.009 | 0.030 | 0.000 | 0.18 | 0.02 | 925.0 | 20.82 | 180.0 |
| 5/15/2025 | 07:06 | -0.001 | 0.007 | 0.000 | 0.000 | 0.075 | 0.030 | 0.000 | 0.000 | 0.000 | 0.000 | 0.090 | 0.000 | 0.000 | 0.000 | 0.005 | 0.000 | 0.031 | 0.000 | 0.08 | 0.03 | 923.0 | 20.79 | 180.0 |
| 5/15/2025 | 07:07 | -0.002 | 0.002 | 0.000 | 0.000 | 0.150 | 0.000 | 0.000 | 0.005 | 0.000 | 0.000 | 0.022 | 0.019 | 0.000 | 0.000 | 0.006 | 0.000 | 0.100 | 0.003 | 0.15 | 0.08 | 922.0 | 20.78 | 180.0 |
| **NO2/Mid O2 Direct** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 07:27 | -0.007 | 0.000 | 0.000 | 0.000 | 98.287 | 0.489 | 0.340 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.017 | 0.000 | 0.000 | 0.025 | 0.000 | 0.000 | 98.29 | 0.00 | 914.0 | 10.52 | 180.0 |
| 5/15/2025 | 07:27 | -0.008 | 0.003 | 0.000 | 0.000 | 98.277 | 0.584 | 0.333 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.017 | 0.000 | 0.000 | 98.28 | 0.00 | 917.0 | 10.52 | 180.0 |
| 5/15/2025 | 07:28 | -0.009 | 0.004 | 0.027 | 0.000 | 98.726 | 0.660 | 0.340 | 0.000 | 0.000 | 0.000 | 0.057 | 0.000 | 0.228 | 0.000 | 0.000 | 0.009 | 0.000 | 0.000 | 98.73 | 0.00 | 916.0 | 10.52 | 180.0 |
| **Spike Direct** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 07:34 | -0.003 | 4.633 | 504.023 | 250.685 | 0.000 | 5.971 | 2.786 | 2.068 | 0.163 | 0.112 | 0.447 | 0.275 | 1.405 | 251.442 | 1.191 | 0.000 | 2.221 | 9.949 | 250.69 | 255.38 | 917.0 | 0.00 | 180.0 |
| 5/15/2025 | 07:34 | -0.004 | 4.649 | 503.908 | 250.872 | 0.000 | 6.284 | 2.838 | 2.096 | 0.314 | 0.155 | 0.481 | 0.273 | 1.650 | 251.293 | 1.155 | 0.000 | 2.249 | 9.942 | 250.87 | 255.21 | 916.0 | 0.00 | 180.0 |
| 5/15/2025 | 07:35 | -0.004 | 4.645 | 504.745 | 250.468 | 0.000 | 6.251 | 2.828 | 2.160 | 0.428 | 0.215 | 0.501 | 0.251 | 2.679 | 251.153 | 0.984 | 0.025 | 2.104 | 9.920 | 250.47 | 254.67 | 918.0 | 0.00 | 180.0 |
| 5/15/2025 | 07:35 | -0.004 | 4.634 | 503.210 | 250.085 | 0.000 | 6.058 | 2.826 | 2.159 | 0.358 | 0.150 | 0.367 | 0.215 | 1.351 | 250.689 | 1.123 | 0.000 | 2.076 | 9.916 | 250.09 | 254.42 | 917.0 | 0.00 | 180.0 |

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **2 Gas Spike Direct** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 07:42 | -0.006 | 0.000 | 0.000 | 0.000 | 0.145 | 0.000 | 0.005 | 1.530 | 0.000 | 0.025 | 0.161 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 99.797 | 10.218 | 0.14 | 66.55 | 909.0 | 0.00 | 180.0 |
| 5/15/2025 | 07:43 | -0.006 | 0.000 | 0.000 | 0.000 | 0.183 | 0.000 | 0.008 | 1.522 | 0.000 | 0.044 | 0.057 | 0.000 | 0.037 | 0.000 | 0.000 | 0.000 | 99.861 | 10.201 | 0.18 | 66.61 | 908.0 | 0.00 | 180.0 |
| 5/15/2025 | 07:43 | -0.005 | 0.000 | 0.000 | 0.000 | 0.160 | 0.000 | 0.007 | 1.509 | 0.000 | 0.038 | 0.080 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 99.516 | 10.182 | 0.16 | 66.37 | 907.0 | 0.00 | 180.0 |
| **Baseline System** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 07:45 | 0.011 | 0.010 | 0.000 | 0.000 | 0.000 | 0.023 | 0.000 | 0.214 | 0.572 | 0.000 | 0.021 | 0.016 | 0.000 | 0.087 | 0.000 | 0.015 | 0.023 | 0.000 | 0.00 | 0.10 | 905.0 | 0.00 | 180.0 |
| 5/15/2025 | 07:46 | 0.009 | 0.016 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.246 | 0.278 | 0.000 | 0.155 | 0.000 | 0.033 | 0.032 | 0.000 | 0.050 | 0.006 | 0.000 | 0.00 | 0.04 | 906.0 | 0.00 | 180.0 |
| 5/15/2025 | 07:46 | 0.003 | 0.010 | 0.000 | 0.000 | 0.237 | 0.139 | 0.000 | 0.275 | 0.192 | 0.000 | 0.000 | 0.000 | 0.028 | 0.011 | 0.000 | 0.040 | 0.022 | 0.000 | 0.24 | 0.03 | 908.0 | 0.00 | 180.0 |
| **CTS System with Response Time** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 07:48 | 0.015 | 0.042 | 0.000 | 2.133 | 0.000 | 0.000 | 0.000 | 0.280 | 0.000 | 100.535 | 0.037 | 0.002 | 0.269 | 0.000 | 0.011 | 0.063 | 0.104 | 0.025 | 2.13 | 67.12 | 913.0 | 0.00 | 180.0 |
| 5/15/2025 | 07:48 | 0.014 | 0.032 | 0.000 | 2.053 | 0.000 | 0.000 | 0.000 | 0.276 | 0.000 | 100.683 | 0.001 | 0.030 | 0.367 | 0.000 | 0.012 | 0.031 | 0.000 | 0.023 | 2.05 | 67.15 | 912.0 | 0.00 | 180.0 |
| 5/15/2025 | 07:48 | 0.011 | 0.036 | 0.000 | 1.780 | 0.000 | 0.000 | 0.000 | 0.192 | 0.000 | 100.876 | 0.000 | 0.000 | 0.447 | 0.000 | 0.000 | 0.063 | 0.217 | 0.023 | 1.78 | 67.40 | 910.0 | 0.00 | 180.0 |
| **NO2/Mid O2 System with Response Time** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 07:56 | -0.000 | 0.023 | 0.990 | 1.078 | 98.837 | 0.491 | 0.355 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.399 | 0.023 | 0.002 | 0.000 | 0.013 | 99.92 | 0.44 | 915.0 | 10.69 | 180.0 |
| 5/15/2025 | 07:56 | -0.000 | 0.021 | 0.951 | 0.913 | 99.139 | 0.402 | 0.353 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.440 | 0.012 | 0.000 | 0.000 | 0.016 | 100.05 | 0.46 | 916.0 | 10.68 | 180.0 |
| 5/15/2025 | 07:56 | -0.000 | 0.026 | 0.963 | 0.797 | 100.015 | 0.444 | 0.360 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.398 | 0.000 | 0.000 | 0.087 | 0.013 | 100.81 | 0.46 | 917.0 | 10.73 | 180.0 |
| **2 Gas Spike System with Response Time** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 07:59 | -0.000 | 0.008 | 0.000 | 0.000 | 0.508 | 0.000 | 0.000 | 1.279 | 0.000 | 0.096 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 97.582 | 10.136 | 0.51 | 65.12 | 919.0 | 0.00 | 180.0 |
| 5/15/2025 | 07:59 | -0.000 | 0.001 | 0.000 | 0.000 | 0.442 | 0.000 | 0.000 | 1.322 | 0.000 | 0.066 | 0.022 | 0.000 | 0.129 | 0.000 | 0.000 | 0.000 | 97.713 | 10.155 | 0.44 | 65.19 | 919.0 | 0.00 | 180.0 |
| 5/15/2025 | 07:59 | 0.001 | 0.000 | 0.000 | 0.000 | 0.141 | 0.000 | 0.000 | 1.360 | 0.000 | 0.063 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 97.964 | 10.174 | 0.14 | 65.35 | 920.0 | 0.00 | 180.0 |
| **Spike System with Response Time** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 08:02 | 0.003 | 4.633 | 510.689 | 251.433 | 0.000 | 6.016 | 2.777 | 1.932 | 1.425 | 0.110 | 0.538 | 0.065 | 0.279 | 252.525 | 1.307 | 0.000 | 2.589 | 9.727 | 251.43 | 256.94 | 921.0 | 0.00 | 180.0 |
| 5/15/2025 | 08:03 | 0.001 | 4.645 | 510.521 | 252.186 | 0.000 | 5.792 | 2.745 | 1.836 | 1.335 | 0.176 | 0.406 | 0.123 | 1.242 | 253.259 | 1.003 | 0.000 | 2.685 | 9.733 | 252.19 | 257.17 | 921.0 | 0.00 | 180.0 |
| 5/15/2025 | 08:03 | 0.002 | 4.653 | 510.547 | 252.043 | 0.000 | 6.004 | 2.739 | 1.844 | 1.095 | 0.157 | 0.325 | 0.044 | 1.109 | 253.629 | 1.003 | 0.000 | 2.703 | 9.734 | 252.04 | 257.54 | 920.0 | 0.00 | 180.0 |
| **Baseline System with Response Time** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 08:06 | -0.000 | 0.011 | 0.542 | 0.000 | 0.000 | 0.055 | 0.000 | 0.013 | 0.000 | 0.000 | 0.021 | 0.000 | 0.000 | 0.195 | 0.000 | 0.013 | 0.132 | 0.006 | 0.00 | 0.28 | 920.0 | 0.00 | 180.0 |
| 5/15/2025 | 08:06 | -0.000 | 0.012 | 0.374 | 0.000 | 0.036 | 0.000 | 0.000 | 0.019 | 0.000 | 0.000 | 0.045 | 0.000 | 0.000 | 0.143 | 0.000 | 0.021 | 0.073 | 0.005 | 0.04 | 0.19 | 920.0 | 0.00 | 180.0 |
| 5/15/2025 | 08:07 | -0.000 | 0.009 | 0.206 | 0.000 | 0.091 | 0.103 | 0.004 | 0.089 | 0.000 | 0.000 | 0.067 | 0.000 | 0.000 | 0.115 | 0.000 | 0.043 | 0.044 | 0.006 | 0.09 | 0.14 | 918.0 | 0.00 | 180.0 |

| Date | Time | $H_2O$ (%) | $CO_2$ (%) | CO (ppmvw) | NO (ppmvw) | $NO_2$ (ppmvw) | $CH_4$ (ppmvw) | $N_2O$ (ppmvw) | $NH_3$ (ppmvw) | $SO_2$ (ppmvw) | $C_2H_4$ (ppmvw) | HCL (ppmvw) | HF (ppmvw) | $C_2H_6$ (ppmvw) | $C_3H_8$ (ppmvw) | $C_6H_{14}$ (ppmvw) | CHOH (ppmvw) | $C_2H_4O$ (ppmvw) | $SF_6$ (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Run 1 - A 30minute/60min** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 10:40 | 19.669 | 9.144 | 3527.211 | 87.556 | 7.703 | 281.126 | 6.792 | 1350.469 | 29.031 | 32.857 | 0.243 | 0.000 | 32.728 | 7.309 | 0.273 | 0.000 | 2.117 | 0.000 | 95.26 | 31.17 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:41 | 19.022 | 9.281 | 3160.098 | 98.164 | 8.277 | 283.862 | 7.433 | 1337.048 | 26.923 | 32.421 | 0.312 | 0.000 | 33.151 | 7.308 | 0.242 | 0.000 | 1.212 | 0.000 | 106.44 | 30.22 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:42 | 18.435 | 9.233 | 3492.856 | 95.776 | 7.980 | 284.338 | 6.981 | 1328.678 | 26.140 | 32.786 | 0.268 | 0.000 | 33.083 | 7.364 | 0.242 | 0.000 | 1.177 | 0.000 | 103.76 | 30.49 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:43 | 18.189 | 9.265 | 3061.766 | 101.147 | 7.409 | 279.745 | 7.582 | 1304.841 | 25.690 | 31.373 | 0.339 | 0.000 | 32.944 | 7.321 | 0.267 | 0.000 | 2.021 | 0.000 | 108.56 | 30.12 | 896.0 | 0.00 | 179.0 |
| 5/15/2025 | 10:44 | 18.136 | 9.287 | 3546.750 | 96.722 | 7.370 | 279.484 | 7.126 | 1322.811 | 24.702 | 33.170 | 0.335 | 0.000 | 32.726 | 7.165 | 0.226 | 0.000 | 1.848 | 0.000 | 104.09 | 30.96 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:45 | 18.079 | 9.269 | 3663.875 | 92.987 | 7.866 | 279.363 | 7.118 | 1328.183 | 23.980 | 33.857 | 0.311 | 0.000 | 32.956 | 7.156 | 0.212 | 0.000 | 1.828 | 0.000 | 100.85 | 31.37 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:46 | 18.126 | 9.299 | 3400.807 | 97.273 | 7.201 | 279.134 | 7.524 | 1317.888 | 24.073 | 32.642 | 0.346 | 0.000 | 33.107 | 7.218 | 0.234 | 0.000 | 1.844 | 0.000 | 104.47 | 30.68 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:47 | 18.126 | 9.258 | 3959.477 | 87.930 | 7.914 | 281.103 | 6.835 | 1331.456 | 24.348 | 34.138 | 0.324 | 0.000 | 33.221 | 7.254 | 0.185 | 0.000 | 1.876 | 0.000 | 95.84 | 31.63 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:48 | 18.133 | 9.300 | 3570.460 | 88.591 | 7.899 | 280.693 | 7.018 | 1311.831 | 24.688 | 32.509 | 0.332 | 0.000 | 33.307 | 7.311 | 0.201 | 0.000 | 1.888 | 0.000 | 96.49 | 30.64 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:49 | 18.144 | 9.278 | 3345.352 | 96.012 | 8.615 | 280.031 | 7.432 | 1321.867 | 24.766 | 33.058 | 0.427 | 0.000 | 33.255 | 7.536 | 0.176 | 0.000 | 1.061 | 0.000 | 104.63 | 30.63 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:50 | 18.068 | 9.208 | 4047.763 | 86.399 | 8.459 | 282.030 | 6.545 | 1329.345 | 25.020 | 33.898 | 0.294 | 0.000 | 33.302 | 7.648 | 0.194 | 0.000 | 1.045 | 0.000 | 94.86 | 31.33 | 897.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:51 | 18.121 | 9.264 | 3241.373 | 98.628 | 8.784 | 279.737 | 7.542 | 1320.005 | 24.970 | 32.495 | 0.294 | 0.000 | 33.011 | 7.833 | 0.197 | 0.000 | 1.140 | 0.000 | 107.41 | 30.65 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:54 | 18.728 | 9.148 | 3647.460 | 97.363 | 8.529 | 279.111 | 7.185 | 1345.796 | 25.903 | 33.960 | 0.364 | 0.000 | 32.820 | 7.808 | 0.211 | 0.000 | 1.119 | 0.000 | 105.89 | 31.62 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:55 | 18.115 | 9.215 | 3666.375 | 99.333 | 8.990 | 283.953 | 7.330 | 1319.614 | 25.263 | 32.885 | 0.335 | 0.000 | 33.040 | 7.855 | 0.191 | 0.000 | 1.080 | 0.000 | 108.32 | 30.88 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:56 | 18.087 | 9.197 | 3686.147 | 97.300 | 8.695 | 283.953 | 7.118 | 1332.522 | 25.446 | 33.734 | 0.273 | 0.000 | 33.417 | 7.860 | 0.169 | 0.000 | 1.025 | 0.000 | 105.99 | 31.37 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:57 | 18.043 | 9.213 | 3599.563 | 101.150 | 9.262 | 284.805 | 7.354 | 1323.421 | 25.730 | 32.993 | 0.374 | 0.000 | 33.373 | 7.859 | 0.175 | 0.000 | 1.122 | 0.000 | 110.41 | 30.95 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:58 | 18.027 | 9.245 | 3229.700 | 110.259 | 9.633 | 285.650 | 7.905 | 1309.137 | 26.063 | 32.018 | 0.322 | 0.000 | 33.462 | 7.724 | 0.163 | 0.000 | 1.031 | 0.000 | 119.89 | 30.08 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 10:59 | 18.050 | 9.218 | 3506.820 | 108.560 | 8.909 | 287.287 | 7.546 | 1311.936 | 26.200 | 32.429 | 0.328 | 0.000 | 33.766 | 7.656 | 0.168 | 0.000 | 1.095 | 0.000 | 117.47 | 30.34 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:00 | 18.911 | 9.104 | 3712.774 | 101.719 | 8.742 | 284.388 | 7.277 | 1333.348 | 26.605 | 32.921 | 0.357 | 0.000 | 32.972 | 7.440 | 0.199 | 0.048 | 1.152 | 0.000 | 110.46 | 30.55 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:01 | 17.988 | 9.246 | 3281.266 | 113.466 | 8.791 | 280.707 | 8.093 | 1310.774 | 25.654 | 31.586 | 0.286 | 0.000 | 32.501 | 7.289 | 0.178 | 0.000 | 1.017 | 0.000 | 122.26 | 29.38 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:02 | 17.955 | 9.257 | 3655.712 | 108.560 | 8.541 | 283.725 | 7.651 | 1311.951 | 25.431 | 32.416 | 0.361 | 0.000 | 32.881 | 7.210 | 0.174 | 0.000 | 0.933 | 0.000 | 117.10 | 29.79 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:03 | 17.969 | 9.256 | 3182.096 | 115.643 | 7.957 | 280.133 | 8.339 | 1301.882 | 25.529 | 31.184 | 0.308 | 0.000 | 32.341 | 7.139 | 0.179 | 0.000 | 0.937 | 0.000 | 123.60 | 28.91 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:04 | 19.036 | 9.070 | 3678.589 | 113.357 | 8.654 | 280.624 | 7.817 | 1338.231 | 25.827 | 33.130 | 0.409 | 0.000 | 32.478 | 7.162 | 0.181 | 0.000 | 1.099 | 0.000 | 122.01 | 30.35 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:05 | 18.200 | 9.217 | 3308.484 | 117.216 | 8.368 | 281.139 | 8.329 | 1311.377 | 25.946 | 31.700 | 0.320 | 0.000 | 32.624 | 7.156 | 0.168 | 0.000 | 1.014 | 0.000 | 125.58 | 29.30 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:06 | 17.979 | 9.995 | 3535.830 | 116.864 | 8.607 | 282.518 | 8.002 | 1320.490 | 25.327 | 32.606 | 0.318 | 0.000 | 32.793 | 7.195 | 0.135 | 0.000 | 0.979 | 0.000 | 125.47 | 29.86 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:07 | 18.759 | 9.113 | 3692.608 | 114.326 | 9.611 | 282.607 | 7.883 | 1315.746 | 26.425 | 31.873 | 0.318 | 0.000 | 32.660 | 7.131 | 0.139 | 0.011 | 1.088 | 0.000 | 123.94 | 29.38 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:08 | 17.861 | 9.245 | 3507.802 | 112.906 | 9.021 | 284.518 | 7.921 | 1306.641 | 25.607 | 31.559 | 0.266 | 0.000 | 33.178 | 7.246 | 0.127 | 0.000 | 0.968 | 0.000 | 121.93 | 29.19 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:09 | 17.843 | 9.977 | 3822.155 | 113.230 | 9.123 | 284.004 | 7.630 | 1313.479 | 25.508 | 32.282 | 0.312 | 0.000 | 32.941 | 7.183 | 0.125 | 0.000 | 0.929 | 0.000 | 122.35 | 29.57 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:10 | 18.068 | 9.218 | 3668.788 | 110.299 | 9.273 | 282.664 | 7.796 | 1304.326 | 25.541 | 31.655 | 0.250 | 0.000 | 32.678 | 7.104 | 0.148 | 0.000 | 1.014 | 0.000 | 119.57 | 29.18 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:11 | 18.463 | 9.132 | 3730.643 | 109.335 | 9.662 | 282.715 | 7.531 | 1334.314 | 25.967 | 32.622 | 0.303 | 0.000 | 32.869 | 7.013 | 0.129 | 0.000 | 1.017 | 0.000 | 119.00 | 29.70 | 896.0 | 0.00 | 180.0 |
| | Avg. | 18.278 | 9.272 | 3537.687 | 102.936 | 8.528 | 282.172 | 7.488 | 1321.647 | 25.610 | 32.625 | 0.321 | 0.000 | 32.986 | 7.382 | 0.187 | 0.002 | 1.256 | 0.000 | 111.463 | 30.343 | 896.03 | 0.000 | 180.0 |

GAS

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Spike 1 - 2-Gas | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 11:15 | 16.300 | 9.147 | 3166.889 | 103.037 | 4.603 | 258.858 | 6.768 | 1162.689 | 25.095 | 20.628 | 0.436 | 0.000 | 27.875 | 6.632 | 0.260 | 0.000 | 11.956 | 0.989 | 107.64 | 28.88 | 896.0 | 0.00 | 180.0 |
| Run 1 - B 30minute/60min | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 11:17 | 17.927 | 9.183 | 3356.882 | 119.153 | 9.410 | 287.428 | 8.015 | 1310.160 | 25.994 | 31.080 | 0.321 | 0.000 | 33.663 | 6.819 | 0.086 | 0.000 | 1.094 | 0.000 | 128.56 | 28.44 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:18 | 18.078 | 9.187 | 3581.428 | 122.309 | 9.501 | 288.091 | 7.971 | 1324.421 | 25.966 | 32.093 | 0.299 | 0.000 | 33.440 | 6.756 | 0.070 | 0.000 | 0.959 | 0.000 | 131.81 | 28.93 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:19 | 18.132 | 9.106 | 4355.859 | 106.094 | 9.933 | 291.789 | 6.875 | 1337.257 | 25.693 | 33.903 | 0.312 | 0.000 | 33.697 | 6.686 | 0.052 | 0.000 | 0.960 | 0.000 | 116.03 | 30.03 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:20 | 18.186 | 9.083 | 4166.352 | 101.226 | 9.993 | 290.627 | 6.924 | 1344.919 | 25.810 | 33.940 | 0.290 | 0.000 | 33.778 | 6.695 | 0.045 | 0.000 | 0.931 | 0.000 | 111.22 | 30.03 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:21 | 19.834 | 8.963 | 3383.796 | 112.559 | 10.155 | 286.824 | 7.982 | 1354.353 | 28.820 | 32.312 | 0.337 | 0.000 | 33.377 | 6.505 | 0.090 | 0.000 | 1.131 | 0.000 | 122.71 | 28.98 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:22 | 18.215 | 9.133 | 3653.242 | 115.035 | 9.589 | 292.240 | 7.744 | 1324.353 | 26.354 | 32.522 | 0.284 | 0.000 | 34.468 | 6.702 | 0.070 | 0.000 | 1.048 | 0.000 | 124.62 | 29.22 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:23 | 18.189 | 9.139 | 3428.898 | 121.413 | 9.955 | 291.456 | 8.112 | 1322.098 | 26.563 | 31.900 | 0.281 | 0.000 | 34.455 | 6.676 | 0.064 | 0.000 | 0.997 | 0.000 | 131.37 | 28.74 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:24 | 18.248 | 9.157 | 3484.996 | 123.586 | 10.337 | 293.096 | 8.127 | 1327.093 | 26.770 | 32.153 | 0.365 | 0.000 | 34.524 | 6.714 | 0.048 | 0.029 | 1.006 | 0.000 | 133.92 | 28.92 | 896.0 | 0.00 | 181.0 |
| 5/15/2025 | 11:25 | 18.161 | 9.133 | 3654.742 | 117.267 | 10.576 | 293.908 | 7.732 | 1310.142 | 27.326 | 31.316 | 0.249 | 0.000 | 34.563 | 6.791 | 0.052 | 0.000 | 1.053 | 0.000 | 127.84 | 28.48 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:26 | 18.071 | 10.388 | 2572.537 | 41.313 | 10.828 | 279.332 | 3.703 | 1545.970 | 24.845 | 29.434 | 0.234 | 0.000 | 31.587 | 5.049 | 0.140 | 0.001 | 1.222 | 0.000 | 52.14 | 25.77 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:27 | 18.149 | 9.033 | 3475.464 | 60.189 | 11.455 | 294.116 | 5.503 | 1404.080 | 27.944 | 33.869 | 0.195 | 0.000 | 34.731 | 6.710 | 0.042 | 0.000 | 1.030 | 0.000 | 71.64 | 30.06 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:28 | 18.174 | 9.049 | 3851.551 | 90.294 | 10.523 | 293.495 | 6.477 | 1374.342 | 26.766 | 34.633 | 0.296 | 0.000 | 34.438 | 6.873 | 0.058 | 0.000 | 0.978 | 0.000 | 100.82 | 30.73 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:30 | 18.212 | 9.109 | 3466.665 | 115.210 | 10.429 | 295.047 | 7.412 | 1354.433 | 27.056 | 33.368 | 0.153 | 0.000 | 34.874 | 6.936 | 0.063 | 0.000 | 0.990 | 0.000 | 125.64 | 29.97 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:31 | 18.148 | 9.090 | 4015.520 | 117.766 | 10.188 | 297.140 | 7.156 | 1339.651 | 26.932 | 33.839 | 0.247 | 0.000 | 34.925 | 6.949 | 0.052 | 0.000 | 0.974 | 0.000 | 127.95 | 30.26 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:32 | 18.120 | 9.095 | 4044.390 | 114.343 | 10.882 | 297.883 | 6.957 | 1349.232 | 26.626 | 34.349 | 0.358 | 0.000 | 35.178 | 6.990 | 0.037 | 0.000 | 0.986 | 0.000 | 125.22 | 30.62 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:33 | 18.160 | 9.161 | 3538.981 | 120.402 | 10.344 | 296.298 | 7.513 | 1330.301 | 26.808 | 32.598 | 0.290 | 0.000 | 35.222 | 6.960 | 0.079 | 0.000 | 0.987 | 0.000 | 130.75 | 29.51 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:34 | 18.107 | 9.160 | 3497.060 | 129.207 | 9.948 | 296.581 | 7.837 | 1319.650 | 26.771 | 32.336 | 0.311 | 0.000 | 35.226 | 6.936 | 0.100 | 0.000 | 0.931 | 0.000 | 139.16 | 29.31 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:35 | 18.120 | 9.181 | 3484.954 | 129.795 | 10.492 | 299.596 | 7.887 | 1312.920 | 26.847 | 31.863 | 0.262 | 0.000 | 35.937 | 7.024 | 0.088 | 0.000 | 0.969 | 0.000 | 140.29 | 29.09 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:36 | 18.129 | 9.126 | 4017.091 | 126.878 | 10.598 | 302.646 | 7.416 | 1327.085 | 26.818 | 33.747 | 0.391 | 0.000 | 36.263 | 7.095 | 0.101 | 0.000 | 0.940 | 0.000 | 137.48 | 30.42 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:37 | 18.187 | 9.099 | 4184.984 | 116.726 | 10.561 | 303.589 | 7.028 | 1348.276 | 26.622 | 34.989 | 0.214 | 0.000 | 36.314 | 7.090 | 0.067 | 0.000 | 0.952 | 0.000 | 127.29 | 31.19 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:38 | 19.444 | 8.977 | 4202.593 | 108.038 | 10.573 | 302.389 | 6.977 | 1352.019 | 29.171 | 34.266 | 0.285 | 0.000 | 36.002 | 6.875 | 0.090 | 0.000 | 1.073 | 0.000 | 118.61 | 30.62 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:39 | 18.337 | 9.067 | 4134.986 | 108.942 | 10.654 | 303.791 | 6.808 | 1358.249 | 28.496 | 35.034 | 0.200 | 0.000 | 36.264 | 6.922 | 0.094 | 0.000 | 0.965 | 0.000 | 119.60 | 31.11 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:40 | 18.257 | 9.148 | 3491.604 | 120.806 | 10.178 | 298.451 | 7.681 | 1331.372 | 26.545 | 32.265 | 0.273 | 0.000 | 35.351 | 6.712 | 0.104 | 0.000 | 0.963 | 0.000 | 130.98 | 29.07 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:41 | 18.249 | 9.146 | 3662.256 | 118.458 | 10.143 | 295.123 | 7.413 | 1319.595 | 26.634 | 31.316 | 0.272 | 0.000 | 34.715 | 6.691 | 0.114 | 0.000 | 1.037 | 0.000 | 128.60 | 28.49 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:42 | 18.255 | 9.168 | 3570.135 | 121.276 | 10.147 | 296.787 | 7.550 | 1305.623 | 26.986 | 30.039 | 0.325 | 0.000 | 35.179 | 6.677 | 0.144 | 0.002 | 1.075 | 0.000 | 131.42 | 27.71 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:43 | 18.271 | 9.152 | 3682.553 | 123.597 | 9.936 | 296.405 | 7.436 | 1327.123 | 26.645 | 31.357 | 0.355 | 0.000 | 34.965 | 6.602 | 0.133 | 0.023 | 0.971 | 0.000 | 133.53 | 28.42 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:44 | 18.352 | 9.157 | 3626.195 | 122.160 | 9.832 | 298.165 | 7.506 | 1329.165 | 26.536 | 31.316 | 0.385 | 0.000 | 35.501 | 6.617 | 0.155 | 0.013 | 0.999 | 0.000 | 131.99 | 28.47 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:45 | 18.305 | 9.072 | 4201.626 | 121.415 | 9.902 | 300.905 | 7.085 | 1346.885 | 27.810 | 33.734 | 0.342 | 0.000 | 35.574 | 6.539 | 0.107 | 0.000 | 0.995 | 0.000 | 131.32 | 29.91 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:46 | 18.327 | 10.005 | 3910.226 | 95.646 | 10.583 | 296.718 | 6.285 | 1377.586 | 26.235 | 33.161 | 0.305 | 0.000 | 34.878 | 6.309 | 0.147 | 0.000 | 1.117 | 0.000 | 106.23 | 29.46 | 896.0 | 0.00 | 180.0 |

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 11:47 | 18.290 | 9.922 | 4140.763 | 105.528 | 10.437 | 299.907 | 6.850 | 1333.387 | 26.632 | 33.524 | 0.352 | 0.000 | 35.064 | 6.435 | 0.123 | 0.000 | 0.989 | 0.000 | 115.97 | 29.69 | 896.0 | 0.00 | 180.0 |
| | Avg. | 18.288 | 9.213 | 3727.944 | 111.554 | 10.269 | 295.327 | 7.199 | 1344.725 | 26.834 | 32.742 | 0.293 | 0.000 | 34.805 | 6.711 | 0.087 | 0.002 | 1.011 | 0.000 | 121.824 | 29.388 | 895.93 | 0.000 | 180.0 |
| | Run 1 Avg: | 18.283 | 9.242 | 3632.815 | 107.245 | 9.399 | 288.750 | 7.343 | 1333.186 | 26.222 | 32.684 | 0.307 | 0.000 | 33.896 | 7.046 | 0.137 | 0.002 | 1.133 | 0.000 | 116.644 | 29.866 | 895.98 | 0.000 | 180.0 |

## Baseline 1

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 11:54 | 0.004 | 0.006 | 0.441 | 0.000 | 0.000 | 0.000 | 0.000 | 2.304 | 0.243 | 0.000 | 0.044 | 0.000 | 0.128 | 0.061 | 0.015 | 0.038 | 0.000 | 0.000 | 0.00 | 0.09 | 896.0 | 0.00 | 180.0 |

## Run 2 - A 30minute/60min

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 11:56 | 17.972 | 9.068 | 3546.277 | 122.563 | 9.895 | 294.601 | 7.417 | 1319.592 | 26.388 | 31.381 | 0.379 | 0.000 | 34.720 | 6.660 | 0.185 | 0.017 | 1.046 | 0.000 | 132.46 | 28.65 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:57 | 18.101 | 9.080 | 4298.503 | 112.533 | 10.934 | 305.748 | 6.867 | 1347.640 | 26.289 | 34.049 | 0.340 | 0.000 | 36.908 | 6.906 | 0.164 | 0.000 | 1.068 | 0.000 | 123.47 | 30.65 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:58 | 18.169 | 9.189 | 3797.651 | 113.672 | 10.159 | 303.536 | 7.342 | 1317.700 | 26.201 | 31.606 | 0.299 | 0.000 | 37.028 | 6.894 | 0.168 | 0.000 | 1.012 | 0.000 | 123.83 | 28.98 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 11:59 | 18.197 | 9.211 | 3657.109 | 122.143 | 10.641 | 302.444 | 7.664 | 1319.116 | 26.284 | 31.374 | 0.460 | 0.000 | 36.676 | 6.902 | 0.204 | 0.000 | 0.989 | 0.000 | 132.78 | 28.89 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:00 | 18.149 | 9.125 | 4571.409 | 108.915 | 10.786 | 304.087 | 6.658 | 1341.806 | 26.109 | 34.114 | 0.319 | 0.000 | 36.729 | 6.913 | 0.183 | 0.000 | 1.044 | 0.000 | 119.70 | 30.72 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:01 | 18.211 | 9.208 | 3517.241 | 119.219 | 10.057 | 298.171 | 7.647 | 1324.767 | 26.139 | 31.313 | 0.320 | 0.000 | 36.131 | 6.762 | 0.219 | 0.000 | 1.106 | 0.000 | 129.28 | 28.81 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:02 | 18.233 | 9.194 | 3775.551 | 123.265 | 9.687 | 298.097 | 7.496 | 1333.388 | 26.315 | 31.929 | 0.324 | 0.000 | 35.972 | 6.787 | 0.192 | 0.000 | 1.096 | 0.000 | 132.95 | 29.19 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:03 | 18.260 | 9.211 | 3862.480 | 116.910 | 9.611 | 297.689 | 7.065 | 1335.027 | 26.426 | 31.801 | 0.427 | 0.000 | 35.789 | 6.748 | 0.175 | 0.008 | 1.703 | 0.000 | 126.52 | 29.44 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:04 | 18.240 | 10.116 | 3711.323 | 110.348 | 9.001 | 291.450 | 6.975 | 1366.773 | 26.046 | 32.024 | 0.280 | 0.000 | 33.565 | 5.965 | 0.222 | 0.000 | 1.723 | 0.000 | 119.35 | 28.91 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:05 | 18.002 | 10.795 | 1567.824 | 32.354 | 4.714 | 274.275 | 2.463 | 1642.399 | 22.788 | 29.924 | 0.243 | 0.000 | 29.261 | 3.751 | 0.416 | 0.000 | 0.877 | 0.000 | 37.07 | 25.12 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:07 | 18.078 | 10.319 | 3092.354 | 38.164 | 9.867 | 289.937 | 4.057 | 1466.484 | 26.080 | 31.935 | 0.362 | 0.000 | 34.424 | 6.013 | 0.237 | 0.000 | 1.462 | 0.000 | 48.03 | 28.75 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:08 | 18.166 | 9.104 | 3377.209 | 62.586 | 9.649 | 292.201 | 5.708 | 1444.238 | 24.695 | 36.568 | 0.352 | 0.000 | 34.963 | 6.715 | 0.196 | 0.000 | 1.045 | 0.000 | 72.23 | 32.19 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:09 | 18.179 | 9.122 | 3621.954 | 88.216 | 9.549 | 294.212 | 6.347 | 1402.431 | 25.390 | 35.476 | 0.354 | 0.000 | 35.371 | 6.958 | 0.197 | 0.004 | 1.096 | 0.000 | 97.77 | 31.73 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:10 | 18.230 | 9.164 | 3363.388 | 111.670 | 9.105 | 292.310 | 7.063 | 1376.071 | 25.933 | 34.104 | 0.286 | 0.000 | 34.836 | 6.921 | 0.235 | 0.000 | 1.045 | 0.000 | 120.78 | 30.82 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:11 | 18.240 | 9.153 | 3720.182 | 118.758 | 9.827 | 296.449 | 6.929 | 1382.033 | 25.773 | 35.215 | 0.337 | 0.000 | 35.893 | 7.057 | 0.215 | 0.015 | 1.105 | 0.000 | 128.59 | 31.70 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:12 | 18.181 | 9.133 | 3986.250 | 113.674 | 9.195 | 294.560 | 6.803 | 1372.216 | 25.799 | 35.218 | 0.260 | 0.000 | 35.313 | 6.939 | 0.230 | 0.000 | 1.040 | 0.000 | 122.87 | 31.57 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:13 | 18.171 | 9.150 | 3846.138 | 113.795 | 8.848 | 295.608 | 6.880 | 1377.254 | 25.671 | 35.134 | 0.399 | 0.000 | 35.744 | 7.080 | 0.267 | 0.000 | 1.035 | 0.000 | 122.64 | 31.73 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:14 | 18.163 | 9.176 | 3781.544 | 114.930 | 9.554 | 295.798 | 7.062 | 1352.870 | 25.740 | 33.489 | 0.305 | 0.000 | 36.074 | 7.169 | 0.245 | 0.000 | 1.069 | 0.000 | 124.48 | 30.70 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:15 | 18.175 | 9.168 | 3998.413 | 115.042 | 9.193 | 296.386 | 6.914 | 1352.941 | 25.796 | 34.348 | 0.470 | 0.000 | 36.094 | 7.233 | 0.246 | 0.006 | 1.072 | 0.000 | 124.24 | 31.34 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:16 | 18.226 | 9.186 | 3797.112 | 116.526 | 9.669 | 295.325 | 7.073 | 1355.715 | 25.789 | 33.854 | 0.354 | 0.000 | 36.037 | 7.291 | 0.253 | 0.000 | 1.075 | 0.000 | 126.19 | 31.08 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:17 | 18.169 | 9.140 | 4730.781 | 99.347 | 9.069 | 297.839 | 6.043 | 1368.860 | 25.794 | 36.233 | 0.291 | 0.000 | 35.685 | 7.236 | 0.249 | 0.000 | 1.070 | 0.000 | 108.42 | 32.60 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:18 | 18.172 | 9.136 | 3925.032 | 105.944 | 8.757 | 293.917 | 6.703 | 1383.431 | 25.229 | 35.147 | 0.350 | 0.000 | 35.539 | 7.323 | 0.281 | 0.000 | 1.089 | 0.000 | 114.70 | 32.04 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:19 | 18.185 | 9.166 | 4436.472 | 103.271 | 9.145 | 297.001 | 6.516 | 1349.822 | 25.754 | 34.170 | 0.385 | 0.000 | 35.834 | 7.370 | 0.269 | 0.000 | 1.071 | 0.000 | 112.42 | 31.40 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:20 | 18.145 | 9.190 | 3600.679 | 117.927 | 9.030 | 292.631 | 7.367 | 1372.376 | 25.031 | 34.392 | 0.365 | 0.000 | 34.906 | 7.163 | 0.284 | 0.000 | 1.032 | 0.000 | 126.96 | 31.35 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:21 | 18.155 | 9.175 | 4245.417 | 110.165 | 9.024 | 296.733 | 6.784 | 1360.946 | 25.320 | 34.745 | 0.350 | 0.000 | 35.775 | 7.268 | 0.279 | 0.000 | 1.059 | 0.000 | 119.19 | 31.70 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:22 | 18.154 | 9.177 | 4195.966 | 102.103 | 9.785 | 296.818 | 6.602 | 1362.284 | 25.499 | 34.526 | 0.290 | 0.000 | 35.234 | 7.131 | 0.272 | 0.000 | 0.966 | 0.000 | 111.89 | 31.34 | 895.0 | 0.00 | 180.0 |

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 12:23 | 18.143 | 9.171 | 3945.169 | 110.144 | 9.159 | 296.712 | 6.996 | 1370.686 | 25.270 | 34.618 | 0.320 | 0.000 | 35.724 | 7.243 | 0.273 | 0.000 | 1.005 | 0.000 | 119.30 | 31.54 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:24 | 18.177 | 9.200 | 3929.694 | 111.932 | 8.996 | 295.439 | 7.139 | 1362.449 | 25.327 | 34.008 | 0.303 | 0.000 | 35.625 | 7.150 | 0.284 | 0.000 | 1.028 | 0.000 | 120.93 | 31.08 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:25 | 18.224 | 9.221 | 3571.414 | 123.228 | 9.402 | 294.924 | 7.616 | 1348.276 | 25.754 | 32.596 | 0.388 | 0.000 | 35.673 | 7.164 | 0.296 | 0.000 | 1.021 | 0.000 | 132.63 | 30.17 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:26 | 18.196 | 9.208 | 3562.276 | 127.783 | 8.608 | 294.937 | 7.683 | 1339.972 | 25.701 | 32.307 | 0.319 | 0.000 | 35.574 | 7.049 | 0.289 | 0.000 | 1.712 | 0.000 | 136.39 | 30.31 | 895.0 | 0.00 | 180.0 |
| | Avg. | 18.169 | 9.289 | 3767.760 | 106.238 | 9.364 | 295.661 | 6.729 | 1371.652 | 25.678 | 33.587 | 0.341 | 0.000 | 35.437 | 6.859 | 0.241 | 0.002 | 1.125 | 0.000 | 115.602 | 30.483 | 895.63 | 0.000 | 180.0 |

## Spike 2 - 2-Gas

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 12:30 | 16.668 | 9.689 | 3304.607 | 109.972 | 6.554 | 278.978 | 6.690 | 1234.216 | 24.894 | 24.356 | 0.384 | 0.000 | 32.190 | 6.701 | 0.252 | 0.000 | 9.033 | 0.639 | 116.53 | 29.47 | 895.0 | 0.00 | 180.0 |

## Run 2 - B 30minute/60min

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 12:32 | 18.202 | 9.138 | 4180.146 | 120.391 | 9.157 | 299.592 | 7.041 | 1368.818 | 25.701 | 34.536 | 0.265 | 0.000 | 35.879 | 6.959 | 0.264 | 0.000 | 1.213 | 0.000 | 129.55 | 31.32 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:33 | 18.237 | 9.206 | 3821.033 | 119.574 | 8.822 | 298.085 | 7.340 | 1354.353 | 25.844 | 32.873 | 0.327 | 0.000 | 35.892 | 6.833 | 0.304 | 0.000 | 0.984 | 0.000 | 128.40 | 30.01 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:34 | 18.119 | 9.125 | 4583.830 | 112.974 | 9.990 | 300.918 | 6.684 | 1365.633 | 25.622 | 35.032 | 0.329 | 0.000 | 35.806 | 6.911 | 0.229 | 0.000 | 1.060 | 0.000 | 122.96 | 31.43 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:35 | 18.164 | 9.116 | 4224.886 | 107.894 | 9.428 | 300.722 | 6.783 | 1371.442 | 25.457 | 34.274 | 0.312 | 0.000 | 36.050 | 6.830 | 0.273 | 0.000 | 0.990 | 0.000 | 117.32 | 30.89 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:36 | 18.165 | 9.148 | 4362.917 | 114.342 | 8.799 | 301.431 | 6.990 | 1355.196 | 25.969 | 33.941 | 0.204 | 0.000 | 35.991 | 6.878 | 0.254 | 0.000 | 1.042 | 0.000 | 123.14 | 30.71 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:37 | 18.199 | 10.161 | 3452.850 | 79.298 | 10.038 | 295.074 | 6.122 | 1402.800 | 25.619 | 30.843 | 0.226 | 0.000 | 35.079 | 6.346 | 0.280 | 0.000 | 1.496 | 0.000 | 89.34 | 28.47 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:38 | 18.181 | 9.141 | 3518.892 | 119.030 | 9.675 | 299.337 | 7.485 | 1377.116 | 25.673 | 33.622 | 0.357 | 0.000 | 35.958 | 6.784 | 0.258 | 0.000 | 1.073 | 0.000 | 128.71 | 30.43 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:39 | 18.204 | 10.033 | 3922.459 | 131.107 | 9.589 | 301.400 | 7.427 | 1362.051 | 25.713 | 33.793 | 0.325 | 0.000 | 36.177 | 6.724 | 0.259 | 0.000 | 0.946 | 0.000 | 140.70 | 30.40 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:41 | 18.200 | 9.149 | 3993.721 | 126.623 | 9.773 | 302.747 | 7.228 | 1361.655 | 25.858 | 33.483 | 0.361 | 0.000 | 36.387 | 6.771 | 0.220 | 0.000 | 1.053 | 0.000 | 136.40 | 30.24 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:42 | 18.235 | 9.189 | 3983.185 | 125.051 | 9.246 | 302.005 | 7.342 | 1357.344 | 25.554 | 32.719 | 0.335 | 0.000 | 36.187 | 6.815 | 0.255 | 0.000 | 0.935 | 0.000 | 134.30 | 29.76 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:43 | 18.188 | 9.166 | 4219.307 | 121.899 | 9.286 | 301.331 | 7.186 | 1355.286 | 25.583 | 33.712 | 0.373 | 0.000 | 35.872 | 6.710 | 0.243 | 0.000 | 1.619 | 0.000 | 131.19 | 30.75 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:44 | 18.221 | 10.011 | 3501.020 | 132.607 | 9.623 | 299.462 | 8.089 | 1339.798 | 25.636 | 31.691 | 0.407 | 0.000 | 35.841 | 6.686 | 0.225 | 0.000 | 1.597 | 0.000 | 142.23 | 29.33 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:45 | 18.197 | 10.017 | 3636.413 | 137.759 | 8.879 | 300.222 | 8.167 | 1340.473 | 25.803 | 32.034 | 0.390 | 0.000 | 35.940 | 6.731 | 0.242 | 0.000 | 1.634 | 0.000 | 146.64 | 29.66 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:46 | 18.186 | 9.195 | 3849.505 | 127.819 | 10.118 | 302.379 | 7.652 | 1334.152 | 25.994 | 31.964 | 0.286 | 0.000 | 36.257 | 6.820 | 0.217 | 0.000 | 1.008 | 0.000 | 137.94 | 29.24 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:47 | 18.178 | 9.191 | 3686.948 | 135.806 | 9.759 | 301.301 | 7.947 | 1341.406 | 26.284 | 32.133 | 0.295 | 0.000 | 36.148 | 6.875 | 0.247 | 0.000 | 1.102 | 0.000 | 145.57 | 29.53 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:48 | 18.210 | 9.159 | 3690.681 | 139.559 | 10.136 | 301.821 | 8.051 | 1348.669 | 26.222 | 32.603 | 0.402 | 0.000 | 35.837 | 6.843 | 0.252 | 0.000 | 0.994 | 0.000 | 149.70 | 29.75 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:49 | 18.165 | 9.141 | 4154.833 | 133.891 | 10.214 | 304.638 | 7.601 | 1341.492 | 26.394 | 33.070 | 0.323 | 0.000 | 36.617 | 7.014 | 0.235 | 0.000 | 1.062 | 0.000 | 144.10 | 30.24 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:50 | 18.150 | 9.185 | 3797.073 | 136.087 | 10.625 | 302.582 | 7.966 | 1342.053 | 26.351 | 32.564 | 0.310 | 0.000 | 36.417 | 7.060 | 0.236 | 0.000 | 1.068 | 0.000 | 146.71 | 29.95 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:51 | 18.130 | 9.169 | 3781.739 | 143.375 | 10.278 | 304.581 | 8.182 | 1347.200 | 26.522 | 33.657 | 0.245 | 0.000 | 36.649 | 7.086 | 0.220 | 0.000 | 1.088 | 0.000 | 153.65 | 30.69 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:52 | 18.114 | 9.168 | 3820.985 | 143.727 | 10.152 | 305.604 | 8.169 | 1342.435 | 26.340 | 33.795 | 0.340 | 0.000 | 36.840 | 7.179 | 0.242 | 0.000 | 1.059 | 0.000 | 153.88 | 30.90 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:53 | 18.124 | 9.169 | 3817.353 | 140.347 | 9.394 | 304.360 | 8.072 | 1345.171 | 26.332 | 33.564 | 0.324 | 0.000 | 36.781 | 7.132 | 0.241 | 0.000 | 0.933 | 0.000 | 149.74 | 30.61 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:54 | 18.146 | 9.182 | 3763.311 | 143.241 | 9.773 | 303.050 | 8.222 | 1345.521 | 26.394 | 32.915 | 0.338 | 0.000 | 36.695 | 7.248 | 0.246 | 0.000 | 1.058 | 0.000 | 153.01 | 30.39 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:55 | 18.132 | 9.148 | 4331.218 | 129.667 | 10.834 | 305.024 | 7.282 | 1349.734 | 26.479 | 33.591 | 0.215 | 0.000 | 37.104 | 7.315 | 0.227 | 0.000 | 1.170 | 0.000 | 140.50 | 30.94 | 895.0 | 0.00 | 180.0 |

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 12:56 | 18.152 | 9.184 | 3744.474 | 136.683 | 10.459 | 302.699 | 8.123 | 1341.939 | 26.656 | 32.346 | 0.324 | 0.000 | 36.818 | 7.293 | 0.259 | 0.000 | 1.054 | 0.000 | 147.14 | 30.08 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:57 | 18.109 | 9.172 | 3826.153 | 140.109 | 9.780 | 302.955 | 8.135 | 1335.100 | 26.547 | 32.697 | 0.366 | 0.000 | 36.740 | 7.324 | 0.254 | 0.000 | 0.873 | 0.000 | 149.89 | 30.21 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:58 | 18.096 | 9.186 | 3918.883 | 137.414 | 9.456 | 304.181 | 7.943 | 1335.932 | 26.241 | 32.408 | 0.342 | 0.000 | 36.877 | 7.268 | 0.255 | 0.000 | 1.080 | 0.000 | 146.87 | 30.11 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 12:59 | 18.070 | 9.159 | 3914.193 | 136.705 | 10.057 | 300.442 | 7.882 | 1347.634 | 26.107 | 33.039 | 0.304 | 0.000 | 36.254 | 7.095 | 0.254 | 0.000 | 1.044 | 0.000 | 146.76 | 30.33 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:00 | 17.923 | 10.597 | 1988.081 | 60.500 | 3.100 | 276.041 | 4.271 | 1561.796 | 22.914 | 31.511 | 0.287 | 0.000 | 28.498 | 4.023 | 0.425 | 0.000 | 1.190 | 0.000 | 63.60 | 26.68 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:01 | 17.971 | 10.372 | 2532.552 | 36.142 | 9.567 | 287.266 | 3.489 | 1481.374 | 25.624 | 26.914 | 0.246 | 0.000 | 33.449 | 5.362 | 0.302 | 0.000 | 1.115 | 0.000 | 45.71 | 24.65 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:02 | 18.050 | 10.223 | 3162.331 | 56.001 | 10.365 | 296.850 | 5.492 | 1430.884 | 25.090 | 33.354 | 0.275 | 0.000 | 35.879 | 6.693 | 0.258 | 0.000 | 1.095 | 0.000 | 66.37 | 30.18 | 895.0 | 0.00 | 180.0 |
| | Avg. | 18.147 | 9.407 | 3772.699 | 120.854 | 9.546 | 300.270 | 7.279 | 1366.149 | 25.884 | 32.823 | 0.314 | 0.000 | 35.897 | 6.787 | 0.256 | 0.000 | 1.121 | 0.000 | 130.401 | 29.929 | 895.00 | 0.000 | 180.0 |
| | Run 2 Avg: | 18.158 | 9.348 | 3770.230 | 113.546 | 9.455 | 297.966 | 7.004 | 1368.900 | 25.781 | 33.205 | 0.328 | 0.000 | 35.667 | 6.823 | 0.248 | 0.001 | 1.123 | 0.000 | 123.001 | 30.206 | 895.32 | 0.000 | 180.0 |
| **Baseline 2** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 13:05 | 0.017 | 0.001 | 1.654 | 0.000 | 0.117 | 0.000 | 0.061 | 4.499 | 0.802 | 0.000 | 0.000 | 0.000 | 0.186 | 0.023 | 0.002 | 0.030 | 0.000 | 0.000 | 0.12 | 0.03 | 895.0 | 0.00 | 180.0 |
| **Run 3 - A 30minute/60min** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 13:07 | 17.759 | 8.966 | 4363.571 | 117.769 | 8.760 | 294.640 | 6.501 | 1334.430 | 25.388 | 32.851 | 0.378 | 0.000 | 34.837 | 7.112 | 0.228 | 0.000 | 0.992 | 0.000 | 126.53 | 30.13 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:08 | 18.091 | 9.090 | 4738.351 | 105.488 | 9.662 | 301.754 | 6.406 | 1347.284 | 27.381 | 34.036 | 0.307 | 0.000 | 36.153 | 7.091 | 0.218 | 0.000 | 1.069 | 0.000 | 115.15 | 30.93 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:09 | 18.165 | 9.113 | 4108.658 | 112.820 | 10.133 | 299.780 | 7.032 | 1352.677 | 25.795 | 33.175 | 0.358 | 0.000 | 36.145 | 7.134 | 0.226 | 0.000 | 0.913 | 0.000 | 122.95 | 30.31 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:10 | 18.202 | 9.139 | 3947.438 | 124.357 | 9.992 | 299.636 | 7.360 | 1360.935 | 25.797 | 33.717 | 0.413 | 0.000 | 36.105 | 7.075 | 0.247 | 0.000 | 1.019 | 0.000 | 134.35 | 30.73 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:11 | 18.192 | 9.182 | 3633.956 | 135.560 | 9.543 | 298.170 | 7.906 | 1354.638 | 25.981 | 32.712 | 0.313 | 0.000 | 35.769 | 6.995 | 0.240 | 0.000 | 1.047 | 0.000 | 145.10 | 29.98 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:12 | 18.258 | 9.189 | 3654.334 | 138.071 | 10.011 | 298.878 | 8.001 | 1352.606 | 26.070 | 32.935 | 0.363 | 0.000 | 35.999 | 7.065 | 0.233 | 0.000 | 0.867 | 0.000 | 148.08 | 30.07 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:14 | 18.260 | 9.236 | 3453.308 | 146.009 | 8.751 | 297.112 | 8.509 | 1343.254 | 26.125 | 31.445 | 0.354 | 0.000 | 35.775 | 7.005 | 0.275 | 0.000 | 1.740 | 0.000 | 154.76 | 29.68 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:15 | 18.239 | 9.214 | 3459.813 | 146.553 | 9.039 | 297.792 | 8.555 | 1337.483 | 26.124 | 30.733 | 0.377 | 0.000 | 35.788 | 7.012 | 0.257 | 0.000 | 1.667 | 0.000 | 155.59 | 29.13 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:16 | 18.217 | 9.215 | 3485.350 | 148.222 | 9.657 | 296.553 | 8.633 | 1332.633 | 26.278 | 30.638 | 0.389 | 0.000 | 35.592 | 7.086 | 0.229 | 0.000 | 1.005 | 0.000 | 157.88 | 28.64 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:17 | 18.186 | 9.220 | 3530.424 | 151.055 | 9.247 | 296.367 | 8.742 | 1334.296 | 26.199 | 30.889 | 0.333 | 0.000 | 35.610 | 7.099 | 0.244 | 0.000 | 1.789 | 0.000 | 160.30 | 29.37 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:18 | 18.169 | 9.219 | 3512.994 | 151.597 | 9.276 | 295.087 | 8.841 | 1328.287 | 25.939 | 30.974 | 0.385 | 0.000 | 35.317 | 7.146 | 0.248 | 0.000 | 1.090 | 0.000 | 160.87 | 29.02 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:19 | 18.143 | 9.231 | 3647.091 | 147.363 | 9.315 | 296.819 | 8.658 | 1327.167 | 25.986 | 31.235 | 0.357 | 0.000 | 35.762 | 7.195 | 0.267 | 0.000 | 0.895 | 0.000 | 156.68 | 29.15 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:20 | 18.122 | 9.215 | 3494.548 | 148.270 | 9.535 | 294.210 | 8.867 | 1326.916 | 25.883 | 30.747 | 0.407 | 0.000 | 35.172 | 7.203 | 0.259 | 0.000 | 0.842 | 0.000 | 157.81 | 28.78 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:21 | 18.124 | 9.199 | 3512.437 | 149.256 | 9.145 | 294.745 | 8.912 | 1331.581 | 25.818 | 30.732 | 0.389 | 0.000 | 35.408 | 7.153 | 0.273 | 0.000 | 0.998 | 0.000 | 158.40 | 28.85 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:22 | 18.134 | 9.206 | 3587.431 | 147.471 | 9.248 | 296.476 | 8.743 | 1333.011 | 26.036 | 30.878 | 0.383 | 0.000 | 35.474 | 7.221 | 0.255 | 0.000 | 0.895 | 0.000 | 156.72 | 28.91 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:23 | 18.118 | 9.197 | 3735.959 | 147.447 | 9.068 | 296.244 | 8.567 | 1341.590 | 26.164 | 31.248 | 0.266 | 0.000 | 35.488 | 7.284 | 0.258 | 0.000 | 1.083 | 0.000 | 156.51 | 29.36 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:24 | 18.071 | 9.193 | 3692.213 | 143.038 | 8.952 | 296.233 | 8.435 | 1339.390 | 26.291 | 31.236 | 0.363 | 0.000 | 35.595 | 7.313 | 0.280 | 0.000 | 1.745 | 0.000 | 151.99 | 29.86 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:25 | 18.033 | 9.170 | 3968.415 | 140.288 | 9.373 | 296.178 | 8.164 | 1345.522 | 25.993 | 31.721 | 0.247 | 0.000 | 35.412 | 7.301 | 0.263 | 0.000 | 1.045 | 0.000 | 149.66 | 29.67 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:26 | 18.090 | 9.185 | 3847.663 | 139.112 | 9.694 | 295.270 | 8.240 | 1349.226 | 25.708 | 31.747 | 0.363 | 0.000 | 35.286 | 7.258 | 0.251 | 0.000 | 1.010 | 0.000 | 148.81 | 29.60 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:27 | 18.075 | 9.229 | 3660.374 | 142.242 | 9.406 | 294.515 | 8.636 | 1343.060 | 26.036 | 31.046 | 0.366 | 0.000 | 35.204 | 7.227 | 0.284 | 0.000 | 1.044 | 0.000 | 151.65 | 29.19 | 895.0 | 0.00 | 180.0 |

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 13:28 | 17.983 | 9.171 | 4014.644 | 139.782 | 8.727 | 294.064 | 8.246 | 1348.439 | 25.459 | 32.024 | 0.401 | 0.000 | 34.992 | 7.223 | 0.284 | 0.000 | 0.957 | 0.000 | 148.51 | 29.78 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:29 | 18.021 | 9.200 | 3808.380 | 136.565 | 8.379 | 294.493 | 8.296 | 1344.150 | 25.593 | 31.395 | 0.435 | 0.000 | 35.093 | 7.184 | 0.278 | 0.000 | 1.492 | 0.000 | 144.94 | 29.67 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:30 | 18.022 | 9.222 | 3566.193 | 137.608 | 9.286 | 295.279 | 8.604 | 1329.283 | 25.596 | 30.098 | 0.390 | 0.000 | 35.314 | 7.145 | 0.258 | 0.000 | 0.964 | 0.000 | 146.89 | 28.37 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:31 | 18.047 | 9.227 | 3647.743 | 140.682 | 9.118 | 293.728 | 8.645 | 1337.341 | 25.587 | 30.590 | 0.384 | 0.000 | 35.039 | 7.011 | 0.290 | 0.000 | 1.019 | 0.000 | 149.80 | 28.66 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:32 | 18.094 | 9.204 | 3717.148 | 136.057 | 9.156 | 294.119 | 8.443 | 1336.466 | 25.572 | 30.561 | 0.387 | 0.000 | 34.901 | 6.950 | 0.249 | 0.000 | 0.970 | 0.000 | 145.21 | 28.47 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:33 | 18.085 | 9.238 | 3484.875 | 142.569 | 9.453 | 294.500 | 8.996 | 1325.725 | 25.773 | 29.790 | 0.437 | 0.000 | 34.952 | 6.878 | 0.250 | 0.000 | 1.594 | 0.000 | 152.02 | 28.30 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:34 | 18.048 | 9.175 | 4179.968 | 134.806 | 9.695 | 299.480 | 8.020 | 1340.785 | 25.886 | 31.847 | 0.332 | 0.000 | 35.653 | 6.926 | 0.245 | 0.000 | 1.024 | 0.000 | 144.50 | 29.33 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:35 | 18.048 | 9.171 | 4123.449 | 122.256 | 9.699 | 298.888 | 7.629 | 1340.460 | 25.839 | 30.852 | 0.337 | 0.000 | 35.936 | 6.948 | 0.286 | 0.000 | 1.013 | 0.000 | 131.96 | 28.76 | 895.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:36 | 18.087 | 9.203 | 3751.557 | 130.281 | 9.607 | 294.263 | 8.151 | 1344.557 | 25.719 | 30.884 | 0.396 | 0.000 | 35.195 | 6.811 | 0.254 | 0.000 | 0.973 | 0.000 | 139.89 | 28.56 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:37 | 18.096 | 9.181 | 3803.176 | 131.551 | 9.045 | 296.334 | 8.171 | 1351.514 | 25.639 | 31.132 | 0.385 | 0.000 | 35.486 | 6.932 | 0.239 | 0.000 | 1.587 | 0.000 | 140.60 | 29.22 | 894.0 | 0.00 | 180.0 |
| | Avg. | 18.106 | 9.187 | 3771.049 | 137.805 | 9.332 | 296.387 | 8.230 | 1340.490 | 25.922 | 31.462 | 0.367 | 0.000 | 35.482 | 7.099 | 0.256 | 0.000 | 1.145 | 0.000 | 147.137 | 29.349 | 894.90 | 0.000 | 180.0 |

## Spike 3 - 2-Gas

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 13:45 | 16.989 | 8.473 | 3271.909 | 123.670 | 5.151 | 276.270 | 7.601 | 1259.339 | 25.156 | 24.173 | 0.431 | 0.000 | 31.102 | 6.887 | 0.474 | 0.000 | 7.410 | 0.471 | 128.82 | 28.89 | 894.0 | 0.00 | 180.0 |

## Run 3 - B 30minute/60min

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 13:48 | 18.037 | 9.103 | 4620.336 | 114.576 | 9.323 | 297.347 | 6.928 | 1377.462 | 25.418 | 33.712 | 0.365 | 0.000 | 35.903 | 7.961 | 0.262 | 0.000 | 1.021 | 0.000 | 123.90 | 31.64 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:49 | 18.088 | 9.159 | 4065.223 | 115.242 | 9.581 | 297.083 | 7.242 | 1363.318 | 25.832 | 31.734 | 0.416 | 0.000 | 36.061 | 8.048 | 0.296 | 0.000 | 0.943 | 0.000 | 124.82 | 30.42 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:50 | 18.140 | 9.187 | 4020.760 | 113.957 | 9.524 | 298.889 | 7.286 | 1360.566 | 26.093 | 31.762 | 0.385 | 0.000 | 36.436 | 8.186 | 0.304 | 0.000 | 0.953 | 0.000 | 123.48 | 30.60 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:51 | 18.137 | 9.193 | 3547.933 | 123.749 | 9.968 | 297.792 | 7.899 | 1352.050 | 26.347 | 30.548 | 0.245 | 0.000 | 36.263 | 8.263 | 0.326 | 0.000 | 1.097 | 0.000 | 133.72 | 30.01 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:52 | 18.166 | 9.189 | 3639.411 | 127.142 | 9.593 | 296.367 | 7.966 | 1351.573 | 25.963 | 30.806 | 0.362 | 0.000 | 36.466 | 8.398 | 0.357 | 0.000 | 0.947 | 0.000 | 136.74 | 30.28 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:53 | 18.144 | 9.213 | 3564.062 | 129.152 | 9.707 | 296.154 | 8.175 | 1348.210 | 35.051 | 30.542 | 0.364 | 0.000 | 36.369 | 8.428 | 0.291 | 0.000 | 1.026 | 0.000 | 138.86 | 30.06 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:54 | 18.166 | 9.165 | 4022.303 | 126.395 | 9.575 | 298.444 | 7.723 | 1373.964 | 35.013 | 32.829 | 0.403 | 0.000 | 36.749 | 8.649 | 0.333 | 0.000 | 1.074 | 0.000 | 135.97 | 31.92 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:55 | 18.141 | 9.193 | 4166.978 | 121.708 | 9.259 | 297.913 | 7.492 | 1369.553 | 35.341 | 32.896 | 0.315 | 0.000 | 36.541 | 8.725 | 0.330 | 0.000 | 0.857 | 0.000 | 130.97 | 31.89 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:56 | 18.085 | 9.205 | 3941.278 | 124.646 | 8.782 | 297.993 | 7.847 | 1354.691 | 35.434 | 32.189 | 0.273 | 0.000 | 36.652 | 8.767 | 0.323 | 0.000 | 1.651 | 0.000 | 133.43 | 31.97 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:57 | 18.014 | 9.125 | 3787.021 | 73.465 | 9.246 | 290.561 | 5.320 | 1448.410 | 24.730 | 30.957 | 0.252 | 0.000 | 35.119 | 7.933 | 0.359 | 0.000 | 1.386 | 0.000 | 82.71 | 30.21 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:58 | 18.102 | 9.184 | 3360.641 | 87.423 | 9.040 | 292.231 | 7.169 | 1397.925 | 35.085 | 32.073 | 0.377 | 0.000 | 35.598 | 8.691 | 0.374 | 0.000 | 1.069 | 0.000 | 96.46 | 31.54 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 13:59 | 18.112 | 9.214 | 3436.677 | 124.679 | 8.801 | 293.586 | 8.083 | 1366.239 | 34.775 | 32.181 | 0.328 | 0.000 | 35.841 | 8.970 | 0.362 | 0.000 | 0.932 | 0.000 | 133.48 | 31.77 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:00 | 18.071 | 9.184 | 3806.257 | 124.728 | 8.382 | 294.473 | 7.538 | 1360.585 | 34.935 | 31.385 | 0.468 | 0.000 | 35.831 | 8.894 | 0.380 | 0.000 | 0.840 | 0.000 | 133.11 | 31.14 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:01 | 18.067 | 9.165 | 4349.918 | 103.038 | 8.902 | 298.983 | 6.832 | 1366.605 | 35.212 | 32.965 | 0.312 | 0.000 | 36.840 | 9.139 | 0.334 | 0.000 | 0.924 | 0.000 | 111.94 | 32.40 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:02 | 18.079 | 9.221 | 3515.178 | 127.774 | 8.589 | 295.845 | 8.044 | 1351.889 | 35.030 | 31.353 | 0.315 | 0.000 | 36.493 | 9.097 | 0.369 | 0.000 | 1.037 | 0.000 | 136.36 | 31.43 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:03 | 18.096 | 9.254 | 3441.048 | 138.354 | 9.140 | 297.739 | 8.462 | 1332.506 | 34.783 | 30.666 | 0.419 | 0.000 | 36.975 | 9.101 | 0.394 | 0.000 | 1.059 | 0.000 | 147.49 | 31.04 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:04 | 18.119 | 9.239 | 3525.927 | 140.391 | 8.493 | 298.024 | 8.398 | 1333.907 | 35.078 | 30.795 | 0.367 | 0.000 | 37.124 | 9.100 | 0.403 | 0.000 | 0.956 | 0.000 | 148.88 | 31.07 | 894.0 | 0.00 | 180.0 |

GAS

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 14:05 | 18.116 | 9.255 | 3494.701 | 143.357 | 9.106 | 294.799 | 8.610 | 1327.523 | 34.891 | 30.661 | 0.242 | 0.000 | 36.222 | 9.114 | 0.390 | 0.000 | 1.016 | 0.000 | 152.46 | 31.01 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:06 | 18.141 | 9.256 | 3521.215 | 145.718 | 8.631 | 295.222 | 8.666 | 1328.881 | 35.252 | 30.898 | 0.425 | 0.000 | 36.550 | 9.129 | 0.416 | 0.000 | 0.976 | 0.000 | 154.35 | 31.21 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:07 | 18.200 | 9.268 | 3600.496 | 145.433 | 8.630 | 295.725 | 8.645 | 1331.334 | 35.017 | 31.234 | 0.288 | 0.000 | 36.326 | 9.247 | 0.384 | 0.000 | 1.034 | 0.000 | 154.06 | 31.53 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:08 | 18.164 | 9.228 | 3821.466 | 141.335 | 8.716 | 297.592 | 8.279 | 1345.879 | 35.219 | 31.742 | 0.352 | 0.000 | 36.637 | 9.282 | 0.394 | 0.000 | 0.989 | 0.000 | 150.05 | 31.89 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:09 | 18.155 | 9.290 | 3543.771 | 143.986 | 8.842 | 294.902 | 8.753 | 1327.476 | 34.975 | 30.217 | 0.439 | 0.000 | 36.148 | 9.235 | 0.403 | 0.000 | 0.878 | 0.000 | 152.83 | 30.77 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:10 | 18.123 | 9.251 | 3833.819 | 140.149 | 8.373 | 297.045 | 8.330 | 1339.487 | 34.866 | 31.545 | 0.269 | 0.000 | 36.855 | 9.316 | 0.390 | 0.000 | 1.042 | 0.000 | 148.52 | 31.82 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:11 | 18.128 | 9.268 | 3691.261 | 142.339 | 8.616 | 296.259 | 8.604 | 1333.646 | 34.942 | 31.186 | 0.349 | 0.000 | 36.578 | 9.304 | 0.435 | 0.000 | 0.934 | 0.000 | 150.95 | 31.59 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:12 | 18.092 | 9.255 | 3712.898 | 139.173 | 8.403 | 295.590 | 8.466 | 1328.467 | 34.670 | 30.753 | 0.351 | 0.000 | 36.428 | 9.413 | 0.405 | 0.000 | 0.974 | 0.000 | 147.58 | 31.38 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:13 | 18.078 | 9.205 | 4236.203 | 133.472 | 8.286 | 296.185 | 7.868 | 1340.363 | 34.855 | 32.081 | 0.355 | 0.000 | 36.574 | 9.427 | 0.413 | 0.000 | 0.987 | 0.000 | 141.76 | 32.30 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:14 | 18.077 | 9.196 | 4495.343 | 120.782 | 7.981 | 295.852 | 7.221 | 1340.289 | 34.888 | 32.334 | 0.347 | 0.000 | 36.418 | 9.434 | 0.407 | 0.000 | 0.865 | 0.000 | 128.76 | 32.38 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:15 | 18.104 | 9.270 | 3565.723 | 128.026 | 7.887 | 293.311 | 8.197 | 1325.865 | 34.363 | 30.214 | 0.292 | 0.000 | 36.298 | 9.319 | 0.433 | 0.000 | 1.046 | 0.000 | 135.91 | 31.03 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:16 | 18.128 | 9.273 | 3598.903 | 140.437 | 8.312 | 291.429 | 8.559 | 1338.392 | 34.642 | 31.168 | 0.426 | 0.000 | 36.156 | 9.378 | 0.452 | 0.000 | 1.025 | 0.000 | 148.75 | 31.75 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:17 | 18.148 | 9.278 | 3597.714 | 145.746 | 7.507 | 287.905 | 8.701 | 1330.725 | 34.332 | 30.450 | 0.381 | 0.000 | 35.504 | 9.453 | 0.453 | 0.000 | 1.040 | 0.000 | 153.25 | 31.35 | 894.0 | 0.00 | 180.0 |
| | Avg. | 18.114 | 9.216 | 3784.149 | 127.546 | 8.840 | 295.708 | 7.910 | 1351.593 | 33.101 | 31.463 | 0.349 | 0.000 | 36.332 | 8.913 | 0.372 | 0.000 | 1.019 | 0.000 | 136.385 | 31.313 | 894.00 | 0.000 | 180.0 |
| | Run 3 Avg: | 18.110 | 9.201 | 3777.599 | 132.675 | 9.086 | 296.047 | 8.070 | 1346.041 | 29.511 | 31.462 | 0.358 | 0.000 | 35.907 | 8.006 | 0.314 | 0.000 | 1.082 | 0.000 | 141.761 | 30.331 | 894.45 | 0.000 | 180.0 |
| **Baseline 3** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 14:25 | -0.000 | 0.020 | 0.123 | 0.000 | 0.063 | 0.000 | 0.003 | 1.554 | 0.347 | 0.000 | 0.021 | 0.000 | 0.114 | 0.049 | 0.036 | 0.063 | 0.000 | 0.000 | 0.06 | 0.12 | 894.0 | 0.00 | 180.0 |
| **Post CTS** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 14:32 | 0.011 | 0.005 | 0.000 | 2.353 | 0.000 | 0.000 | 0.000 | 1.096 | 0.000 | 100.410 | 0.056 | 0.003 | 0.600 | 0.000 | 0.045 | 0.130 | 0.226 | 0.029 | 2.35 | 67.18 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:32 | 0.022 | 0.308 | 17.652 | 0.000 | 0.000 | 0.000 | 1.896 | 0.000 | 0.000 | 98.914 | 0.000 | 1.095 | 0.000 | 0.000 | 0.000 | 2.861 | 7.515 | 0.000 | 0.00 | 70.96 | 894.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:32 | 0.011 | 0.011 | 0.000 | 2.447 | 0.000 | 0.000 | 0.000 | 1.030 | 0.000 | 99.971 | 0.000 | 0.000 | 0.384 | 0.000 | 0.023 | 0.000 | 0.191 | 0.029 | 2.45 | 66.82 | 894.0 | 0.00 | 180.0 |
| **Post NO2/Mid O2** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 14:36 | -0.000 | 0.002 | 0.092 | 3.124 | 96.189 | 0.383 | 0.364 | 0.388 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.040 | 0.003 | 99.31 | 0.03 | 894.0 | 10.53 | 180.0 |
| 5/15/2025 | 14:36 | -0.000 | 0.000 | 0.000 | 2.920 | 97.073 | 0.000 | 0.361 | 0.220 | 0.000 | 0.000 | 0.000 | 0.001 | 0.000 | 0.000 | 0.000 | 0.000 | 0.207 | 0.000 | 99.99 | 0.14 | 894.0 | 10.43 | 180.0 |
| 5/15/2025 | 14:37 | 0.002 | 0.000 | 0.000 | 2.635 | 96.681 | 0.313 | 0.353 | 0.374 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.002 | 0.000 | 0.000 | 0.000 | 99.32 | 0.00 | 894.0 | 10.55 | 180.0 |
| 5/15/2025 | 14:37 | -0.000 | 0.001 | 0.269 | 2.410 | 96.764 | 0.026 | 0.376 | 0.029 | 0.000 | 0.000 | 0.047 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.137 | 0.001 | 99.17 | 0.09 | 895.0 | 10.58 | 180.0 |
| **Post 2-Gas (Acetaldehyde)** | | | | | | | | | | | | | | | | | | | | | | | | |
| 5/15/2025 | 14:39 | -0.000 | 0.036 | 0.000 | 0.000 | 0.164 | 0.000 | 0.000 | 1.888 | 0.000 | 0.048 | 0.000 | 0.000 | 0.009 | 0.000 | 0.000 | 0.000 | 97.809 | 10.260 | 0.16 | 65.24 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:39 | 0.001 | 0.030 | 0.000 | 0.000 | 0.924 | 0.000 | 0.000 | 1.928 | 0.000 | 0.024 | 0.000 | 0.004 | 0.421 | 0.000 | 0.000 | 0.000 | 98.721 | 10.261 | 0.92 | 65.83 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:40 | 0.004 | 0.030 | 0.000 | 0.000 | 0.000 | 0.000 | 0.008 | 2.327 | 0.000 | 0.001 | 0.231 | 0.000 | 0.037 | 0.000 | 0.000 | 0.000 | 98.725 | 10.246 | 0.00 | 65.82 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:40 | 0.002 | 0.030 | 0.000 | 0.000 | 0.542 | 0.000 | 0.000 | 1.916 | 0.000 | 0.035 | 0.000 | 0.000 | 0.028 | 0.000 | 0.000 | 0.000 | 99.082 | 10.225 | 0.54 | 66.08 | 896.0 | 0.00 | 180.0 |

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 14:40 | 0.005 | 0.036 | 0.000 | 0.000 | 0.303 | 0.000 | 0.021 | 1.841 | 0.000 | 0.093 | 0.000 | 0.011 | 0.000 | 0.000 | 0.000 | 0.000 | 99.365 | 10.181 | 0.30 | 66.31 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:40 | -0.000 | 0.033 | 0.000 | 0.000 | 0.335 | 0.005 | 0.000 | 1.790 | 0.000 | 0.057 | 0.000 | 0.000 | 0.230 | 0.000 | 0.000 | 0.000 | 99.725 | 10.189 | 0.33 | 66.53 | 896.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:40 | 0.003 | 0.000 | 0.000 | 0.000 | 0.439 | 0.049 | 0.038 | 2.130 | 0.000 | 0.076 | 0.000 | 0.000 | 0.038 | 0.000 | 0.000 | 0.000 | 99.459 | 10.160 | 0.44 | 66.36 | 896.0 | 0.00 | 180.0 |

## Post 5-Gas

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 14:43 | 0.004 | 4.633 | 499.932 | 249.865 | 0.000 | 5.323 | 2.760 | 2.509 | 2.892 | 0.293 | 0.179 | 0.054 | 1.698 | 252.726 | 0.971 | 0.000 | 2.417 | 9.693 | 249.87 | 256.47 | 897.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:43 | -0.000 | 4.621 | 497.354 | 249.367 | 0.000 | 5.859 | 2.701 | 2.753 | 1.860 | 0.160 | 0.432 | 0.106 | 1.070 | 251.834 | 1.224 | 0.000 | 2.599 | 9.721 | 249.37 | 256.12 | 897.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:44 | 0.005 | 4.628 | 498.327 | 249.657 | 0.000 | 5.807 | 2.796 | 2.655 | 1.730 | 0.146 | 0.307 | 0.136 | 0.391 | 252.051 | 1.369 | 0.000 | 2.669 | 9.700 | 249.66 | 256.67 | 897.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:44 | -0.000 | 4.632 | 498.822 | 249.546 | 0.000 | 6.061 | 2.754 | 2.702 | 1.487 | 0.186 | 0.244 | 0.036 | 1.451 | 252.330 | 1.145 | 0.000 | 2.835 | 9.710 | 249.55 | 256.63 | 897.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:44 | -0.000 | 4.630 | 499.097 | 249.726 | 0.000 | 5.671 | 2.753 | 2.207 | 2.329 | 0.260 | 0.208 | 0.078 | 1.785 | 252.318 | 1.105 | 0.000 | 2.652 | 9.700 | 249.73 | 256.47 | 897.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:44 | 0.005 | 4.623 | 498.602 | 250.095 | 0.000 | 5.726 | 2.718 | 2.623 | 2.588 | 0.216 | 0.152 | 0.094 | 1.569 | 251.902 | 1.151 | 0.000 | 2.857 | 9.692 | 250.09 | 256.25 | 898.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:44 | 0.001 | 4.632 | 497.759 | 249.421 | 0.000 | 5.391 | 2.754 | 2.508 | 1.922 | 0.225 | 0.083 | 0.057 | 1.316 | 252.061 | 1.088 | 0.000 | 2.364 | 9.678 | 249.42 | 255.96 | 898.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:44 | -0.000 | 4.647 | 498.979 | 250.091 | 0.000 | 5.647 | 2.738 | 2.430 | 2.097 | 0.342 | 0.336 | 0.012 | 2.498 | 252.518 | 0.929 | 0.000 | 2.843 | 9.662 | 250.09 | 256.50 | 898.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:44 | 0.002 | 4.645 | 498.150 | 250.208 | 0.000 | 5.781 | 2.744 | 2.487 | 2.528 | 0.354 | 0.506 | 0.087 | 2.377 | 252.335 | 1.046 | 0.000 | 2.588 | 9.669 | 250.21 | 256.39 | 898.0 | 0.00 | 180.0 |

## End of Test Purge

| Date | Time | H2O (%) | CO2 (%) | CO (ppmvw) | NO (ppmvw) | NO2 (ppmvw) | CH4 (ppmvw) | N2O (ppmvw) | NH3 (ppmvw) | SO2 (ppmvw) | C2H4 (ppmvw) | HCL (ppmvw) | HF (ppmvw) | C2H6 (ppmvw) | C3H8 (ppmvw) | C6H14 (ppmvw) | CHOH (ppmvw) | C2H4O (ppmvw) | SF6 (ppmvw) | Nox (ppmvw) | VOC (ppmvw) | Ambient Press. (mbar) | Oxygen (%) | Cell Temp (C). |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5/15/2025 | 14:48 | -0.000 | 0.000 | 0.322 | 0.000 | 0.000 | 0.156 | 0.030 | 0.400 | 0.000 | 0.000 | 0.246 | 0.037 | 0.221 | 0.133 | 0.000 | 0.000 | 0.206 | 0.000 | 0.00 | 0.27 | 900.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:49 | -0.000 | 0.001 | 0.367 | 0.000 | 0.450 | 0.113 | 0.018 | 0.404 | 0.000 | 0.000 | 0.000 | 0.000 | 0.006 | 0.111 | 0.005 | 0.062 | 0.182 | 0.007 | 0.45 | 0.24 | 900.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:49 | -0.000 | 0.005 | 0.302 | 0.089 | 0.000 | 0.000 | 0.000 | 0.539 | 0.000 | 0.000 | 0.000 | 0.000 | 0.230 | 0.086 | 0.000 | 0.112 | 0.177 | 0.009 | 0.09 | 0.20 | 900.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:49 | -0.000 | 0.005 | 0.315 | 0.000 | 0.402 | 0.000 | 0.000 | 0.246 | 0.000 | 0.058 | 0.000 | 0.009 | 0.199 | 0.120 | 0.014 | 0.021 | 0.082 | 0.000 | 0.40 | 0.24 | 900.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:49 | -0.000 | 0.001 | 0.324 | 0.000 | 0.053 | 0.000 | 0.008 | 0.399 | 0.000 | 0.088 | 0.000 | 0.000 | 0.021 | 0.026 | 0.027 | 0.025 | 0.000 | 0.002 | 0.05 | 0.14 | 900.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:49 | -0.000 | 0.000 | 0.173 | 0.000 | 0.036 | 0.000 | 0.023 | 0.343 | 0.000 | 0.110 | 0.000 | 0.021 | 0.033 | 0.007 | 0.051 | 0.000 | 0.035 | 0.004 | 0.04 | 0.21 | 900.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:49 | -0.000 | 0.001 | 0.190 | 0.000 | 0.000 | 0.000 | 0.008 | 0.403 | 0.000 | 0.000 | 0.097 | 0.014 | 0.140 | 0.098 | 0.013 | 0.106 | 0.230 | 0.002 | 0.00 | 0.28 | 900.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:49 | -0.000 | 0.003 | 0.267 | 0.000 | 0.033 | 0.000 | 0.009 | 0.414 | 0.000 | 0.000 | 0.175 | 0.000 | 0.138 | 0.084 | 0.030 | 0.032 | 0.000 | 0.003 | 0.03 | 0.14 | 900.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:49 | -0.000 | 0.000 | 0.302 | 0.000 | 0.000 | 0.000 | 0.011 | 0.491 | 0.000 | 0.000 | 0.110 | 0.022 | 0.150 | 0.047 | 0.040 | 0.117 | 0.069 | 0.000 | 0.00 | 0.17 | 900.0 | 0.00 | 180.0 |
| 5/15/2025 | 14:49 | -0.000 | 0.006 | 0.261 | 0.000 | 0.000 | 0.000 | 0.012 | 0.343 | 0.037 | 0.000 | 0.079 | 0.002 | 0.000 | 0.015 | 0.002 | 0.081 | 0.000 | 0.004 | 0.00 | 0.02 | 900.0 | 0.00 | 180.0 |